

Project Submission Pro-Forma

Student name: Vishnu Prashanth Sridhar

Student ID: 5569451

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Signed: Vishnu Prashanth Sridhar

Date: 03-09-2024



**A Blockchain-Enabled Evaluator of Indicators of Compromise in Connected
Vehicles**

by

Vishnu Prashanth Sridhar

Dissertation submitted in partial fulfilment for the Degree of Master of Science in Cyber Security
Engineering

WMG
University of Warwick

Submitted September 2024

Declaration

I have read and understood the rules on cheating, plagiarism and appropriate referencing as outlined in my handbook and I declare that the work contained in this assignment is my own, unless otherwise acknowledged.

No substantial part of the work submitted here has also been submitted by me in other assessments for this or previous degree courses, and I acknowledge that if this has been done an appropriate reduction in the mark I might otherwise have received will be made.

Project definition for my degree (as copied from
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The project for MSc Cyber Security Engineering must:

1. address a research question directly relating to cyber security, AND
2. demonstrate understanding of the particular issues around conducting research in the cyber domain, AND
3. conduct research in the cyber domain in an appropriate manner.

My project relates to this definition in the following way:

My individual research project directly addresses the critical research question within the cybersecurity field on investigating how blockchain technology can assist in mitigating DDoS attacks on CANs within CAVs. This individual project represents serious knowledge of specific problems at play within the cyber domain, particularly those issues inherent in securing real-time vehicle communication networks from such complex cyber threats.

This work reflects critically on the unique cybersecurity implications of deploying a decentralized solution, such as blockchain, within resource-constrained environments such as CAN networks, while scalability, latency, and resource demands are considerations. To that effect, conducting the research in a controlled and simulated environment will ensure that the investigation is appropriately conducted with methodological rigor and considering ethical implications against applying these kinds of technologies in real-world automotive systems.

This approach fully meets the academic requirements for an MSc in Cyber Security Engineering and constitutes a truly valuable contribution to research into how to enhance the resilience and security of critical infrastructure within the cyber environment.

Abstract

The research work focuses on the blockchain research in the detection and mitigation strategies against DDoS attacks on Controller Area Networks in Connected Autonomous Vehicles. Because CAVs are developing rapidly and increasingly dependent on complicated inter-vehicular and intra-vehicular networks, including CAN as they introduce significant cyber-physical threats, especially DDoS attacks against critical vehicle operations. Blockchain is a decentralised and transparent immutable framework. The research investigates how blockchain would increase the security of CAN networks in CAVs and their resilience against such an attack.

This study falls under a positivist paradigm through quantitative research for representing the feasibility study and effectiveness of DDoS mitigation using blockchain. A simulation was designed and then used in the study, with a virtual CAN environment employed in CAVs to subjectively test the efficiency of Blockchain in controlled conditions to detect and mitigate DDoS attacks. Results have proved that blockchain enhances mitigation and detects DDoS attacks by eliminating single failure points while maintaining the integrity of communication in the network. In contrast, this research shows that several challenges about scalability and latency in blockchain with their high resource requirements should be overcome so that their implementation does not adversely impact network performance.

Some of the major contributions of this thesis are to develop a blockchain-based detection and mitigation system and provide some practical recommendations on optimizing blockchain for latency reduction and less resource consumption. These findings represent the statement that blockchain has potential for application in the improvement of CAV cybersecurity. In this regard, it also underlines the need for further research to be conducted in refining the proposed solution and ensuring their practical applicability for real world scenarios . Future research directions concern investigating the integration of artificial intelligence with blockchain to further enhance the predictive capabilities of security frameworks in CAVs.

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CyBok Area

The research presented in this thesis primarily falls within the "**Network Security**" and "**Security Operations and Incident Management**" areas of the Cyber Security Body of Knowledge (CyBOK). It addresses critical aspects of securing communication networks in Connected Autonomous Vehicles (CAVs) against Distributed Denial-of-Service (DDoS) attacks by leveraging blockchain technology, which encompasses network security principles and threat intelligence sharing to ensure data integrity, resilience, and effective incident management. The inclusion of Security Operations emphasizes the importance of continuous monitoring, threat detection, and response strategies in maintaining the security of autonomous vehicular networks.

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Chapter 1 Introduction

1.1 Introduction

The research and development in the automotive sector have given rise to Connected Autonomous Vehicles (CAVs), which are connected through inter-vehicular and intra-vehicular networks such as Controller Area Networks (CAN) and the Internet, particularly in the network and transport layers of the OSI model, exposes CAVs to a range of cybersecurity threats, including Distributed Denial of Service (DDoS) attacks. This thesis explores how a blockchain-enabled system for evaluating Indicators of Compromise (IoCs) can serve as an effective detection and mitigation strategy against DDoS attacks targeting CAN networks in CAVs. The guiding research question is, “How effective blockchain can be utilized for DDoS mitigation in CAV Controller Area Networks (CANs), addressing scalability, latency, and security challenges”. This study discusses the advantages of blockchain compared to other methods and its effectiveness, which could increase the security of CAV networks. The aim of the thesis is to highlight the potential of blockchain and the distributed ledger technology in enhancing the reliability and accessibility of CAV's.

Further, the transparency characteristic of blockchain technology increases the trust among various stakeholders in the automotive infrastructure, as all transactions in the blockchain and numerous associated nodes are recorded immutably, allowing for real-time auditing between the vehicles and infrastructure. This feature enhances operational integrity but also supports collaboration among various participants in the CAV ecosystem, paving way for innovative solutions and shared advancements. Also, as the technology evolves, the potential for decentralized applications to further streamline processes and improve user experiences becomes a priority driving the decentralisation adoption of CAV systems across the sector. As organizations recognise the benefits, investment in research and development will likely increase, leading to extensive research that could redefine the industry.

1.1.1 Overview of the thesis

The advancements in the automotive sector technology have paved the way for the development of connected autonomous vehicles that will revolutionise the future transportation systems. Such vehicles will be able to work independently and intercommunicate with other vehicles and infrastructure, creating a smart network with the enhancement for the safety, efficiency, and

convenience . CAVs can drive through sophisticated scenarios by using advanced sensors, Artificial Intelligence (AI), and other communication systems to make prompt decisions and collaborate with systems like traffic control centres and other vehicles to avoid human failure. Further, as the technology is constantly evolving in relation to CAVs, the adaptation within existing transportation systems. Provides this backbone for the operation of these vehicles is the CAN, which allows the ECUs of the vehicle(s) to communicate and run critical operations such as steering, braking, acceleration, and engine performance, these ECUs make the CAN system critical to ensure that the vehicle is reliable. Even though it plays a significant role, the CAN does not comply with security by design and privacy by design principles, therefore they are vulnerable to the cyber-attacks and threats. From the various types of attacks , DDoS attack presents significant threat to the CAVs as it can flood the CAN with a lot of data, then compromise the ECU communications and result in critical system failures and eventually loss of vehicle control (Lakhal *et al.*, 2020).

The cybersecurity risks to CAVs evolve with their development, given their growing digitalisation and connectivity. Because this makes them the most vulnerable to malicious actors, successful cyber-attacks can have very serious effects on a CAV, including the risk to passengers' lives and other road users. Hence, well it emerges as the reason why cybersecurity must top among the issues the manufacturers, regulators, and researchers should focus on. Traditional cybersecurity techniques are insufficient for the handling of exegesis of the modern DDoS attack in the dynamic and constrained CAV environment. Many of such solutions centralise, which in turn builds a single point of failure or a new possible target for attacks. This, in turn, may add unacceptable latency to the central service, which can afford no delay for vehicle operation in real time. Therefore, with increasing number of CAVs the need for an effective, resilient, and scalable solution to cybersecurity grows. Blockchain technology has emerged as a promising answer to these challenges, with its decentralized, transparent, and immutable framework, for enhancing the security (Gupta et al., 2020).

As the blockchain system works on a distributed network, it does not have any single point of failure, giving it a high resiliency to DDoS attacks as opposed to an ordinary, centralized system.

Also, the immutability of the blockchain records is also relevant, because once a cyber threat is detected and information is recorded, the record cannot be tampered with, hence providing a valid basis for managing the attack and conducting post-incident analyses. Nevertheless, there can be a high level of usefulness in performing the process of implementing blockchain in CAV cybersecurity, which is at a very primary phase, with big challenges to face: such as scalability of the blockchain

network, latency of the block validation process, and the heavyweight of computational load by running a blockchain-based system over the scarce computing resources available in a vehicle. It evaluates the possibility of implementing blockchain for the security advancement of CAVs, which would offer an approach to mitigate DDoS attacks on the CAN.

The thesis also explores the various blockchain architectures that, can be applied to automotive applications, which have their strengths and weaknesses with real-time data processing examined, along with their resilience for the developing cyber threats. Also assess the integration of smart contracts in such a way that security protocols will be automatically executed when an anomaly is detected, and it is timely responded to for the maintenance of system integrity and performance. This will also establish whether the impact of regulatory compliance and standardization within the automotive sector serves as impetus toward the widespread adoption of blockchain technologies. This comprehensive approach will provide an enhanced foundational base for improving security in connected and autonomous vehicles, hence leading to an improved cyber-safe transport system.

1.1.2 Aim and objectives of the research

The main objectives of this research are to develop and evaluate a blockchain-based solution to assess the various indicators of compromise, with a specific focus on the detection and mitigation of DDoS attacks in CAN for CAV's ,aiming to enhance the security of the controller area network.

1. Identifying the specific challenges associated with implementing blockchain-based solutions for DDoS mitigation in CANs within CAVs.
2. Assessing the feasibility of integrating blockchain technology into existing DDoS mitigation strategies for CANs in CAVs, considering factors such as scalability, latency, and resource constraints.
3. Investigating the potential benefits and drawbacks of employing blockchain for DDoS mitigation in CAVs, including an analysis of the effectiveness of blockchain-based solutions in preventing and mitigating DDoS attacks.
4. Developing solutions or recommendations to address the identified challenges and enhance the effectiveness of blockchain-enabled DDoS mitigation in CANs of CAVs, including potential improvements or enhancements needed for practical implementation in real-world scenarios.

By achieving these objectives, this study will improve the security and resilience in CAVs through the integration of blockchain-based solutions for DDoS detection and mitigation.

1.1.3 Summary of contributions

This paper presents a blockchain-based solution for the detection and mitigation of DDoS attacks against the CAN in CAV's. This study contributes to the domain of cybersecurity in connected autonomous vehicles, proposing a blockchain-based solution for detecting and mitigating distributed denial-of-service attacks against the controller area network in autonomous vehicles.

The research aims to address security vulnerabilities in Connected and Autonomous Vehicles (CAVs), particularly within their controller area networks (CAN), by leveraging data exchange for detecting indicators of compromise (IoCs) and assessing threats. It seeks to develop a simulation testbed integrating security frameworks to proactively mitigate evolving cyber risks in CAV environments.

1.This thesis reviews the various methods for the security against CAN attacks on connected autonomous vehicles. It allows for an in-depth assessment of the different strengths, weaknesses, and suitability to protect critical vehicle networks from DDoS attacks.

2.The thesis proposes a novel blockchain-based system to identify various IoCs present in the CAN of CAV's by using blockchain technology for infrastructure and data to enhance the accuracy of detection and security as it has immutable and decentralised characteristics.

3.Creating a testbed to simulate real-world scenarios, this platform will propose an environment that simulates CAN network in Connected and Autonomous Vehicles (CAV) and simulate Distributed Denial of Service (DDoS) attacks. It will also allow for the assessment of the proposed blockchain solution within the simulated isolated conditions.

4.The analysis of a blockchain-based DDoS mitigation solution with respect to its performance compared to other techniques in critical metrics about performance evaluation, including detection accuracy, response time, scalability, and resource efficiency. This observation showcased the feasibility and effectiveness in practice of blockchain technologies securing CAV networks.

These contributions could make the research effective for enhancing CAV cybersecurity, for a more secure and resilient vehicle system while developing the adoption of blockchain technology within the automotive industry.

1.1.4 Layout of the thesis

Chapter 1: Introduction: This chapter provides an overview of the research, overview of the thesis, background, aim of the research, research objectives, and summary of contributions. And it also outlines the layout of the thesis.

Chapter 2: Literature Review: This chapter presents a review of the existing literature for CAN in Connected and Autonomous Vehicles (CAVs), focusing on DDoS attacks and blockchain technology. The chapter identifies gaps in current research that this thesis will address.

Chapter 3: Methodology: This chapter details the research methodology, including the design and development of the blockchain-based solution. It outlines the system and simulation design, methods for data collection, and analysis methods used in the study.

Chapter 4: Results and Analysis: This chapter presents the results from the experimental evaluation of the proposed solution. It analyses the performance of the blockchain-enabled system in detecting and mitigating DDoS attacks within the CAN network of CAVs

Chapter 5: Discussion : This chapter discusses the implications of the findings, recommendations for practice, and the contribution of the research to the body of knowledge. It also addresses the limitations of the study and suggests areas for future research.

Chapter 6: Conclusions and Future Work: This chapter summarizes the key findings, discusses the significance of the achievements, and provides recommendations for further research. The chapter concludes with a reflection on the research objectives and the extent to which they have been achieved.

1.2 Background

1.2.1 Security Issues in CAN networks in Connected Autonomous Vehicles (CAVs)

CAN is a bus protocol designed for maintain the reliability in in-vehicle communications, V2V and V2X but it suggests weak security features due to the hardware resource constraints within the ECUs, which limit them from performing complex cryptographic operations and functions the simple methods of encryption can be potentially cracked or breached by the attackers who use more computational resources.(Aliwa et al., 2021)

(Bozdal, Samie and Jennions, 2018) discuss about the various security concerns, issues, and mitigation techniques in the Controller Area Network (CAN) bus protocol, which is widely used in the CAV industry for communication between electronic control units (ECUs) in V2V and V2I networks. The CAN protocol which is developed based on the principles such as bit stuffing, error handling, and message priority arbitration, lacks encryption and authentication methods, failing to meet the Confidentiality, Integrity, and Availability (CIA) triad which is essential for system security (Petit and Shladover, 2015). The author categorises the CAN network attacks into three main types: eavesdropping, data insertion, and denial of service (DoS).

CAN's exposed broadcast nature makes it vulnerable to eavesdropping, allowing adversaries to intercept sensitive vehicle data and manipulate critical vehicle functions like the engine, brakes, and airbags by injecting or altering data frames. (Koscher et al., 2010).

DoS and DDoS attacks can disrupt CAN networks by methods of bit overwriting, message flooding, or exploiting error mechanisms which potentially disrupt the network, making critical resources unavailable for various vehicular communications by targeting interfaces such as Ethernet, Bluetooth, and over-the-air updates (Rawat, Bajracharya and Yan, 2015; Bozdal, Samie and Jennions, 2018).

(Checkoway et al., 2011) states that DoS and DDoS attacks pose a significant threat to CAVs, as they can severely impact vehicle safety and functionality affecting the critical functions of the vehicle's ECU when the CAN network is compromised. Possible Indicators of Compromise (IoCs) for these attacks include sudden spikes in network traffic, unexpected message priorities, and anomalous error frame generation.

The existing research propose methods which include network segmentation, encryption, authentication techniques, and network intrusion detection and prevention systems (Groza et al., 2012). But implementing these methods and frameworks face challenges because of CAN's limited bandwidth, processing power, and robust nature. (Bozdal, Samie and Jennions, 2018) emphasize the need for cost-effective and compatible solutions that balance security with existing vehicle architectures. They conclude that continuous research is essential to enhance the robustness of CAN networks against evolving threats in the context of CAVs (Hamad, Nolte and Prevelakis, 2017).

1.2.2 Various Cyber-Attacks in CAV

(Petit and Shladover, 2015) discuss the various cyber-attacks posed in the CAV's and their impact on Connected and Autonomous Vehicles (CAVs) networks. Technology hold immense potential for

transforming transportation systems with improved safety and efficiency. However, their reliance on open wireless communication and internet connectivity makes them susceptible to various cyber-attacks that can undermine these benefits. (Balaji, Almani and Fumell, 2023) This research investigates the impact of four major types of attacks namely Distributed Denial of Service (DDoS), Man-in-the-Middle (MITM), Blackhole, and Grayhole on the network and performance of CAV networks in transportation.

According to this paper among the attacks, the DDoS attack emerged as the most severe attack which causes decrease in network availability and reliability. By flooding the network with volume of packets from multiple simulated attacker nodes. This severe degradation of the network's capacity to transmit valid messages could disrupt critical safety communications and rerouting functions, compromising the operational efficiency and safety of the entire CAV system.

1.2.3 Cyber Security posture in Connected Autonomous Vehicles

(Sun, Yu and Zhang, 2022) provides a comprehensive overview of the posture of cyber security in CAV's which highlights the increasing threat of cyber-attacks due to the enhanced connectivity and automation in autonomous vehicles. The survey categorises cyber-security risks into in-vehicle network attacks, V2X network attacks, and other types of attacks. This details various defence and mitigation strategies and discusses existing cyber-security and safety standards relevant to CAVs. Further, it identifies challenges and problems in the field, also suggests that addressing these issues is crucial for the safety and efficiency of CAVs. This emphasises the importance develop strategies to mitigate unspecified security threats and comply with important security by design and privacy by design principles in cybersecurity standards while designing vehicular networks and CAV's this paper helps us to identify and learn about the various cyber security postures present in the CAV environment.

1.2.4 DDoS attacks and impact on CAN in CAVs

A DDoS attack is a security incident within a network or system where the host faces unexpected data traffic which may be false requests or malicious traffic, which disrupts communication and thus degrades performance of the vehicle and infrastructure. In such an event on the CAN network, performed by a threat actor or attacker typically in this case, interrupting communications between ECUs. Which may result in serious failures, up to failures of whole control systems or car control, which is a critical risk for passenger safety and integrity. The vulnerability to these forms of attack

grows in proportion to the increased number and degree of digitisation and connectedness of CAVs; therefore, CAV cybersecurity measures against these potential threats.(Al-Dhuraibi and Elhadeif, 2021)

1.2.5 Distributed Ledger and its applications in CAN in CAV

The deployment of distributed ledger technology in connected autonomous vehicles provides a probable solution to mitigate cybersecurity risks and threats. Its decentralised, transparent, and immutability characteristics improves the security in CAVs by improving the security against cyberattacks. The distributed network of blockchain has no single point of failure. It is therefore less likely to be vulnerable to DDoS attacks because the recorded parameter cannot be altered after it is stored. Essentially, the technique has a tremendous deal of potential to lock down CAN in CAVs, which would increase system's resilience, threat detection, and data integrity.(Baldini et al., 2020)

1.2.6 Challenges in implementing blockchain based DDoS mitigation in CANs of CAVs

Blockchain and Distributed ledger can improve CAV security, but there are a few challenges to overcome before it can be effectively used to identify and dissolve DDoS attacks in CANs. Scalability of the blockchain network is a significant issue that might affect the system's capacity to effectively manage high volumes of network traffic. The block validation delay could have an impact on CAV systems' real-time performance criteria which may affect the availability of the vehicle or system. Further, it is necessary to solve the computing demands of operating a blockchain-based system in the resource-constrained context of a vehicle. Excluding these difficulties, blockchain's advantages such as its immutability and decentralised feature make it a useful mechanism for enhancing CAV security. The purpose of this study is to examine these issues , address these challenges and look at methods to effectively use blockchain technology for mitigating DDoS attacks in CAVs, contributing to the development of more secure and resilient transportation systems.(Dargahi et al., 2021).

Chapter 2 Literature Review

2.1 Introduction

This literature review discusses the various aspects of CAV cybersecurity, providing comprehensive approaches on cyber threats and specific attack vectors such as DDoS attacks, using blockchain as a detection and mitigation strategy and develop novel threat intelligence sharing system. The analysis focuses on the benefits of blockchain against traditional strategies and as a mitigation strategy and detection mechanism within the CAV networks. This is to underpin the potential role played by blockchain in enhancing the integrity and availability of CAV ecosystems.

2.2 Review of DDoS Attacks in CAN in CAV's

2.2.1 Methods and Types of DDoS Attacks

According to (Beitollahi and Deconinck, 2012) Distributed Denial of Service attacks form a very vulnerable and challenging category of cyber threat notably within the complex systems like Connected Autonomous Vehicles. Essentially, the aim of a DDoS attack is to disrupt the target system from being able to offer any intended service by consuming all available resources, which could include bandwidth, processing capability, or memory. On successful attack the intended and legitimate users are denied access to these critical facilities which decreases the reputation and reliability of the system.

It is therefore imperative to have a thorough understanding of the various forms and methods utilised to carry out DDoS attacks, specifically under the concept of CAVs and their associated communication networks, to design the most effective approaches for detection and mitigation (Douligeris and Mitrokotsa, 2004). The DDoS attacks can be classified based on their volume, attack vectors, distribution, duration, target specificity, and impact on real-time systems. Understanding these characteristics is crucial for developing effective mitigation strategies.

Volumetric Attacks is the most common type of DDoS attack, where an attacker consumes a target's bandwidth by sending excess volumes of data. The attack attempts to saturate the network resources of the targeted or victim system, hence reducing their ability to handle genuine traffic. These types of attacks include UDP floods, ICMP floods, and DNS amplification attacks (Beitollahi and Deconinck, 2012) but not limited. Therefore, in context to Connected and Autonomous Vehicles, volumetric attacks can overload the communication systems onboard the vehicle and cause the denial of service that will prevent the V2V and V2I messages transmissions.

Attack vectors There are three ways which DDoS attacks can be classified into volumetric attacks, protocol-based attacks, and application-layer attacks

Protocol Attacks are attacks which specifically target the vulnerabilities which are existing within the network protocols. For example, SYN floods represent attacks performed against the three-way handshake mechanism implemented in TCP.(Dulik, 2019).In certain cases, the attacker floods the target system with connection requests that never complete the handshake protocol. Due to this, resource exhaustion is caused, because the system is left in an intermediate state, constantly expecting a response to a request that never exists. Other examples include fragmented packet attacks , ping of death, which are based on sending specifically constructed packets to the target to be exploited in the target system's network flow management. Protocol-related attacks can be extremely dangerous to Connected and Autonomous Vehicles since they have an impact on the very core of the communications between the various ECUs within the vehicle's CAN.

Application-layer attacks are conducted at a higher operational layer and directly involve the applications or services of the targeted system. Instead of sending huge volumes of traffic, request patterns like those of normal users are sent by the attacker but at a magnitude which is beyond the capacity of the target application to handle. Application-layer attacks that most commonly overwhelm such web services, rendering them unavailable, include the following: HTTP GET/POST floods.(Black and Kim, 2022)

In the context of CAVs, if such attacks would be launched against onboard vital systems, for instance, navigation or emergency braking services, with a view to causing a catastrophic event. These different types of DDoS attacks will have different effects on the network and communication system architecture in CAVs. It means that the greatest challenge for security researchers is not only to identify such breaches in a very short time but also to mitigate their impact without disturbing the functional efficiency of the vehicle.

2.2.2 Impact of DDoS attacks on CANs in CAVs.

The impact caused by distributed denial of service attacks on connected autonomous vehicles and their internal networks, such as the CAN, are definitely significant . This is because these networks are vital to vehicle control and communication . CAVs are systems that depend on continuous information exchange and data transfer between devices such as onboard sensors, actuators, and

external infrastructure, such as road-side units. Inaccessibility to any of these communication systems due to the DDoS attack will have severe effects, from traffic interruptions to major safety hazards .(Yadav, Ansar and Chaurasia, 2022)

The CAN is a vital communication protocol which vehicles that interconnects the many Electronic Control Units (ECUs) responsible for controlling critical vehicle functionality, such as engine control, braking, steering, and airbags (King and Yu, 2017). Primarily proposed for in-vehicle communication, the CAN now forms part of the CAV, which expects raw data to be shared in real time within and outside the vehicle operating safely in various conditions. For instance, the CAN protocol is not designed with some integral security mechanisms like encryption or authentication. Which makes them exposed to cyber risks and threats, including denial of service and distributed denial of service attacks, that can disrupt the functions of the vehicle.

DoS or DDoS attacks on the CAN networks within the CAVs, can result in delayed or loss of messages, and therefore, it is impossible to process the critical commands to get directed to their target device or system in time . For instance, if a DDoS attack flooding the CAN bus with high-priority messages might cause a delay in the delivery of a no-brake command, which results in delayed or no braking at all. In view of the real-time communication requirement for autonomous decision-making, any delay in data transmission can trigger poor vehicle performance or even dangerous conditions like accidents.(Gupta et al., 2014; Kaur, Kumar and Bhandari, 2017)

In conventional context, the DDoS attack to the CAV network is very likely to disrupt vehicle-to-everything V2X communication. This will not only obstruct the attacked vehicle but also disrupt the traffic systems, forcing accidents, and delayed emergency responses. Further, if several CAVs are attacked simultaneously, the effect on the intelligent transportation system could be such that a catastrophic network disruption and malfunction results in possibly starting traffic deadlocks or even causing system wide outages (Khan et al., 2020).

Due to the critical role that the CAN protocol plays in the operations of CAVs, there is a requirement to follow certain security-by-design mitigation techniques that are not only effective but also efficient in this limited dynamic environment. The CAN networks experience limited processing power and bandwidth; thus, the implementation of typical DDoS mitigation techniques can be a bit challenging, which demands the need for novel techniques using blockchain technology which have privacy-by-design and security-by-design characteristics.

2.3. Current Detection and Mitigation Strategies in CAV's

2.3.1 Existing techniques for DDoS detection and mitigation in CAV's

Defending against DDoS attacks requires a hybrid approach of detection mechanisms with mitigation strategies that are designed specifically for the architecture of CAVs. Conventional DDoS attack detection techniques and mitigation strategies have advanced considerably in recent years to include advanced machine learning algorithms, signature-based techniques in detection, and hybrid methodologies that use more than one technique to perform an attack on a targeted system.

Signature-based detection

Signature-based detection identifies the known attack patterns or signatures. This is among the oldest and most common method used for detecting DDoS attacks. Signature-based systems match the coming network traffic against a database or repository of known attack patterns, flagging any traffic that matches the defined signature pattern. This approach has the advantage of fast detection of known threat with minimal false positive rate (FPR). However, it is not effective against constantly evolving threats, as the attackers might bypass detection based on signatures by varying their attack patterns. In the case of DDoS attacks against CAVs, the rate of evolution of attacks that specifically target the vehicle communications is quite fast, making the signature-based detection scheme outdated (Gupta et al., 2014; Kaur, Kumar and Bhandari, 2017).

Anomaly-Based Detection

In anomaly detection method, threats are detected by determining a variation from normal behaviour of the traffic to an anomalous or abnormal behaviour of traffic. These systems are especially appropriate for an environment such as CAVs, where to start with, the patterns in network traffic are somewhat predictable under normal conditions. Machine learning algorithms can also enhance anomaly-based detection by analysing large amounts of network data to set a baseline for normal traffic and then detecting subtle deviations which might indicate a DDoS attack. Signatures do not exist for zero-day attacks. However, these systems are effective in operation only to the extent that an accurate baseline has been previously established and high-quality training data has been provided for machine learning models (Alkasassbeh, 2018; Alfardus and Rawat, 2024).

Hybrid-Approach

In hybrid approach the DDoS detection systems integrate the signature-based and anomaly-based methods to increase the detection rate and accuracy. Such hybrid methods combine the signature-based detection for known threats and use anomaly-based detection method for unknown and novel attacks. Such hybrid method distinctively defends against a much comprehensive range of DDoS attack vectors therefore it is quite effective in complex and dynamic CAV environments. And the outcome of the study concluded that the hybrid intrusion detection system is efficient than the conventional approaches.(Cepheli, Büyükçorak and Karabulut Kurt, 2016).

Mitigation Strategies:

Mitigation usually includes the use of firewalls, traffic filtering, and rate limiting. Firewalls and traffic filtering systems prevent the malicious traffic from making its way to critical systems, while rate limiting breaks originating traffic from a source from overloading a network with too much traffic. Another traffic absorption technique is load-balancing, wherein the traffic is distributed over multiple servers to prevent any single point failure in the network.

Packet filtering firewalls are efficient firewalls in terms of resource utilization, which impose restrictions on data exchange even with authenticated external devices in case those external devices might be compromised. Firewalls at network gateways further control access to the internal network as a secondary defence mechanism. Firewalls and filtering of traffic function as the primary defence against DDoS attacks, which block or prevent the incoming anomalous and malicious traffic based on predefined rules and detect abnormal data traffic patterns. But with the nature of these high-volume attacks, the efficiency of the traditional methods of firewalls is very low, particularly in the constrained CAN network environments which is present in CAVs. So, more sophisticated strategies are necessary.(Sahana, Gotkhindikar and Tiwari, 2022)

Rate limiting sets the upper limit on the incoming request rate. This ensures that no source can flood any system with traffic. Thus, it mitigates volumetric DDoS attacks. This kind of method would be helpful with CAVs, and other vehicles fitted with specific software or hardware components to protect the critical communication against the overflow of malicious or abnormal simulated traffic so that critical commands are allowed for transmission.(Yogesh Patil and Ragha, 2011)

Load balancing and High Availability proxy refers to the process of distributing traffic coming in across different nodes in the mentioned system or servers in our case it is CAV such that no single system is exploited and is made unavailable. That would ensure the communication of the vehicle with its

external environment is not disrupted due to sudden outbursts of traffic. However, load balancing in a CAV network can be quite challenging with respect to resource constrained CAN architecture and dynamic CAV environments.(Zebari et al., 2020)

Advanced Mitigation Strategies SDN and NFV:

In advanced mitigation methods the use of Software-Defined Networking (SDN) and Network Function Virtualisation (NFV) over these years has been prevalent against DDoS attacks in various network layer level mitigation especially in CAV networks. As they are much more flexible, with greater control over network traffic, and provides for a more improved detection and mitigation strategy.

SDN allows for a centralised control over the traffic in a manner that any DDoS attack can be easily highlighted and mitigated without affecting the availability of the system. SDN provides a more dynamic and responsive approach for managing the network by separating the control and data planes. For example, SDN can change the traffic flows instantly in occurrence of a distinct attack using machine learning algorithms.(Ahuja et al., 2021)

In autonomous vehicles, the in-vehicle network (IVN) or CAN bus network enables data transmission among sensors, electronic control units, and remote servers to provide data processing and management. But, in case of a DDoS attack the compromised sensors once gained access can flood the network with excessive packets, causing packet loss and transmission delays that compromise driving safety. To address this issue the new method proposed which was using Joint K-means clustering and Software-defined networking (JKS) which detects and removes distributed denial of service (DDoS) attacks in the vehicle. By leveraging SDN's centralized control and dynamic response capabilities, JKS efficiently mitigates DDoS attacks by rerouting traffic or removing the compromised device from the network ensuring the integrity of critical data transmission is complete. (Huang, Huang and Chen, 2022)

Network Function Virtualisation (NFV) virtualises network functions so that they can be deployed and managed in software rather than hardware. To this end, flexibly programmable resource scheduling and resource management can be implemented to make it easy for network operators to respond to a DDoS attack. For example, NFV can be leveraged for the CAV network to dynamically allocate resources to critical communication channels during an attack so that these can keep running.(Jakaria et al., 2016)

These mitigation methods hold great potential in advanced networks but at the same time raise a few challenges for the implementation. Resource constraints in CAV networks do not utilize the available remediation techniques effectively. Further, there is a little implementation complexity of the centralized solutions based on the literature with the dynamic nature of CAVs.

2.3.2 Application and limitations in the context of CANs in CAVs.

While a significant number of such traditional strategies for DDoS mitigation, like firewalls or traffic filtering, rate limiting, and load balancing, have been applied sufficiently in several different network environments, their application directly to the context of Controller Area Networks in CAVs proves to have a different set of unique challenges. Where originally the whole design for the communications inside a vehicle, CAN networks based on constrained processing and bandwidth with little available storage, the current resource-intensive mitigation mechanisms would prove to be unsuitable.

Resource constraints are one of the major challenges in applying traditional DDoS mitigation techniques to CAN networks. Unlike traditional computer networks, which have a good amount of processing power and storage capacity, CAN networks are designed to be lightweight and efficient, hence inducing minimal overhead. That makes it very challenging to run resource-intensive mitigation techniques like deep packet inspection or machine learning-based anomaly detection.

Communication faces another challenge in terms of real-time requirements in CAVs. A lot of critical vehicle functions, like braking, steering, and acceleration, are time-based and rely on real-time communication between ECUs. Any form of holdup or disruption in this communication may strongly jeopardize the security of the vehicle. Traditional DDoS mitigation techniques generally require inspection and filtering of traffic that introduces latency unacceptable in the context of CAN.

There is lack of security features in CAN as it doesn't have security features like encryption, authentication, and integrity checking, so it is very vulnerable to cyber-attacks. Although traditional methods to mitigate DDoS attacks might be useful in protecting the network from external threats, they do very little to address the intrinsic vulnerabilities of the CAN protocol itself.

These points give the currently evident realization that, while the threat of DDoS attacks continues in CAVs, the conventional problems of DDoS mitigation techniques themselves cannot individually assure the security of CAN networks. Rather, new and innovative approaches are needed in considering the unique constraints and requirements for CAN networks.

2.4 Blockchain as DDoS detection and mitigation strategy

2.4.1 Overview of blockchain principles as a mitigation and detection strategy in CAV

(Taghavi Zargar et al., 2013) outlines an approach to mitigate Distributed Denial of Service (DDoS) attack, highlighting three primary phases such as prevention, detection, and response. Prevention involves securing systems and protocols by implementing robust security measures, regularly updating software, and ensuring fail-safe protection through service replication. Key strategies include resource allocation, reconfiguration of network topologies, and deploying firewalls and Intrusion Detection and Prevention Systems (IDPS) to block unauthorised access, have proper access control lists and filter attack flows. Additionally, load balancing and network flow control mechanisms are utilised to distribute traffic evenly and prevent server overload and address the various vulnerabilities at the application level.

During an attack, detection mechanisms are very essential for identifying malicious traffic and anomalous patterns, with the most active detection points being at the victim's side, fast and earlier detection near the source of attack is more effective in prevention. After the attack, response mechanisms focus on identifying the source and blocking malicious traffic, collaborating with various vendors and stakeholders to implement measures to minimise the impact of attack and ensure continuous service availability. This multi-layered and directed defence and mitigation strategy aims to effectively tackle the risks and impacts of DDoS attacks through complete prevention, timely detection, and managed response efforts.

The paper presents an architecture which combines blockchain and smart contracts to enable a collaborative DDoS mitigation across multiple levels of the communication such as Application and Network layer . The main advantages of adopting the smart contracts and distributed ledgers in this architecture is the use of a public and distributed blockchain infrastructure to share threat or attack information about IP addresses which can be a potential DDoS source and implement mitigation rules, without the need for specialised registries by using the advantage of immutability the data can be tamper-proof and transparent . The system enables the sharing of DDoS attack information in a fully distributed and automated method using blockchain and smart contracts.(Rodrigues et al., 2017)

Blockchain can be used as an intrusion detection system (Blockchain Assisted Intrusion detection system-BIDS) designed to enhance the security and integrity of Connected and Autonomous Vehicle (CAV) networks, addressing the critical issue of Distributed Denial of Service (DDoS) attacks.

Considering our research case it starts with the Controller Area Network (CAN), which is foundational for vehicle communication systems. The BIDS framework recommended leverages blockchain technology combined with machine learning (ML) techniques within a 5G-V2X communication network to detect and mitigate cyber threats efficiently. The decentralised nature of blockchain ensures secure, immutable data handling and trust management, significantly enhancing the resilience of CAV networks. In the discussed and proposed system, Autonomous Vehicles (AVs) communicate through various technologies like DSRC, mmWave, and 5G-V2X, with Roadside Units (RSUs) facilitating data storage and trust information exchange. (Anbalagan et al., 2023)

The blockchain employs a Proof-of-Work (PoW) consensus mechanism to validate transactions, preventing data tampering and ensuring integrity of data frames inbound-outbound data from the CAN network of individual vehicles. The machine learning component detects and classifies malicious vehicles in real time, using historical and current data to identify anomalies and signatures which are then categorised into various attacks such as DoS and other attacks which does not include DDoS. Trust management between nodes is integral to BIDS, where each node continuously evaluates the trustworthiness of peers, and this information is securely recorded on the blockchain. This mechanism helps isolate malicious vehicles swiftly known as insider nodes or malicious nodes. Constant performance evaluations through various simulations demonstrate the system's effectiveness in reducing inactivity and enhancing accuracy of intrusion detection. (Sani et al., 2024)

2.4.2 Current applications of blockchain and distributed ledger in CAN - CAV's

In recent studies, several have proposed certain proof-of-concept to enhance the security and resilience against DDoS attacks targeting CAV networks. The most promising application of blockchain technology in CAVs is to enable secure V2X (vehicle-to-everything) communication.

An extensive possible approach of blockchain would be creating a fully decentralised and trust less communication framework, enabling secure and duly authenticated message transmission between vehicles and the infrastructure, thereby removing the necessity for a central authority. Particularly, in a blockchain-based V2X communication system, the verification of each vehicle's validation of identity can be conducted through cryptographic methods, ensuring that only vehicles with proper authorisation have access to the network.

Introducing AIT an AI-enabled Trust Management system and the integration of blockchain technology has also been examined for securing intra-vehicular communication among CAVs. For

example, studies have suggested that employing blockchain to validate the authenticity of reports concerning untrustworthy vehicles and to revoke their security credentials, ensuring that malicious vehicles are effectively identified, and their credentials are cancelled. The AIT system demonstrates better performance compared to the existing methods in accurately and efficiently managing trust and detecting malicious vehicles or nodes. This development mitigates the risk of fraudulent message injection and DDoS attacks, enhancing the overall security and reliability of vehicular communication networks.(Zhang et al., 2021). The above discussed method mitigates the risk of malicious nodes or vehicles introducing manipulated messages into the network or executing DDoS attacks.

Another domain where blockchain technology can be instrumental within CAVs pertains to shared data broadcasting and collaborative efforts. CAVs generate data which can be shared with other vehicles and infrastructure to optimise traffic management, by enhancing overall system safety. This interaction sequence creates a significant challenge the secure and reliable sharing of data among CAVs and the linked infrastructure. Blockchain can efficiently address this challenge by providing a secure and transparent platform for data sharing. Therefore, blockchain can securely allow tamper-proof data sharing among vehicles, thus mitigating and addressing the vulnerabilities to various types of attacks and other cyber threats.(Rajendar et al., 2023)

2.4.3 Current Research on blockchain for DDoS mitigation

Intrusion Detection System in CAV

(Lokman, Othman and Abu-Bakar, 2019) deployment of Intrusion Detection Systems (IDS) within CAV systems plays a vital role in end point security in autonomous vehicles, the appropriate placement of IDS in the Electronic Control Units (ECUs) for Host-based IDS, and NIDS on the Controller Area Network (CAN) for network-based monitoring, and at network gateways for intrusion detection in the network layer of CAV's. Specifically, the anomaly-based IDS is crucial due to the predictable nature of CAN traffic, categorising them into frequency-based, machine learning-based, statistical-based, and hybrid-based approaches, each using detailed techniques, blockchain and machine learning algorithms to identify abnormalities from normal network behaviour. The study also discusses about the technical challenges in implementing IDS in vehicles, such as the limited computational resources of ECUs, the need for real-time processing to prevent data transmission delays, and the different traffic patterns of CAN networks, advocating for the integration of IDS with lightweight cryptographic measures to enhance security against attacks.

Malicious node detection in Blockchain

To address the issue of insider nodes (Ajayi and Saadawi, 2021), they have proposed an architecture and system to identify and prevent malicious activities from compromised insider blockchain nodes. This approach leverages smart contracts to authenticate and detect malicious nodes by continuously examining their behaviour. The study focuses mainly on three types of attacks such as attempts to compromise the blockchain, inject false data, or retrieve unauthorised data. The evaluation shows that the architecture effectively detects and isolates insider threats by monitoring detection times also studies indicate this method's advantage in detecting both external and insider nodes, making it a robust solution for maintaining security in blockchain and individual nodes.

Malicious contract identification in Blockchain

The paper (Kang et al., 2023) discusses various methods for detecting malicious smart contracts on the Ethereum blockchain. It highlights the MAIAN tool, which uses symbolic analysis to detect greedy, suicidal, and prodigal contracts by analysing bytecode and performing concrete validation on a corrupted or fake Ethereum network. The paper also analyses other detection systems, that identifies malicious users before they can deploy contracts which converts solidity code to XML for XPath-based vulnerability detection. Additionally, it introduces, a deep learning-based system that evaluates smart contracts before and after deployment, ensuring secure execution on the Ethereum network by identifying greedy contracts through frequency analysis.

Threat intelligence sharing using Blockchain

(Chatziamanetoglou and Rantos, 2024) examines the integration of blockchain technology with cyber threat intelligence (CTI), focusing on the collection, storage, analysis, and sharing of CTI in a secure and trusted approach using blockchain on the communication phase, giving importance to communication and sharing mechanisms across various nodes and in between blockchains. This also observes a development towards strategic and technical goals , with technologies like smart contracts, machine learning, and encryption being essential roles. Trust and privacy are identified as main areas, with the reliability of the integrated systems layer early to mid-phases of development. The potential area of further research is this model or study can be implemented in CAV's to share threat intelligence of malicious attacks of insider nodes or external attacks.

2.5 Identified Research gaps

After conducting a critical review of the literature, several research gaps have been identified, which this thesis aims to address:

1. Despite the advancements in security provided by blockchain technology, its scalability, along with latency, remains very critical in resource-constrained environments such as CAN networks in CAVs. From the literature review, there is a lack of solutions that effectively balance security and real-time operational requirements of autonomous vehicles.
2. There is a noticeable gap in the literature on integrating of blockchain with traditional DDoS mitigation strategies. The potential for blockchain to complement these methods without introducing excessive overheads or complexity needs further evaluation.
3. Most of the existing research is limited to theoretical models or simulations and has not taken much consideration on real-world applications and testing. Understanding how blockchain-based solutions perform in live CAV environments, particularly concerning network latency and overall system performance, remains as an new area of research.
4. The outcomes of implementing blockchain in CAVs, including cost-benefit analyses and the practical challenges of common implementation methods, are not sufficiently discussed in the current literature. This is a critical area, especially for assessing the feasibility of such technologies in the automotive industry.

2.6 Section Conclusion

The review of the literature and the study establishes the security challenges associated with DDoS attacks against CAVs and the potential remedy through blockchain. In contrast to the existing DDoS detection and mitigation strategies, which are not quite feasible in CAV scenarios, blockchain comes out with a possibility of an alternative that can assure security, integrity, and resilience for CAV networks. This calls for further research in order to solve the various challenges and limitations of these blockchain-based approaches in particularly within the resource-constrained environments of CAN networks in CAVs.

Chapter 3 Methodology

3.1 Introduction

Identified through the review of literature and existing methods. Specifically, the research by (Chaganti, Bhushan and Ravi, 2023) highlights that there is a need for the design of a more scalable and secure DDoS detection and mitigation strategy. This method provides the model for the proposed

research “How can blockchain can be used as a potential detection and mitigation strategy against DDoS attacks in CAN for CAVs?” and research objectives To evaluate the feasibility of utilising blockchain technology for DDoS mitigation, identify and analyse the challenges associated with integrating blockchain into DDoS mitigation strategies, assess the effectiveness of blockchain-based solutions in preventing and mitigating DDoS attacks, explore potential needed for blockchain-based DDoS detection and mitigation techniques , to provide insights into the practical implementation of blockchain for DDoS mitigation in real-world case scenarios and sharing threat intelligence using a blockchain enabled framework.

3.1.1 Philosophical Paradigm

In any research, it is essential to identify and justify the philosophical assumptions that influence the approach. These assumptions provide the base for choosing specific methods and techniques. This research accepts a positivist paradigm, grounded in the belief that understanding of network anomalies and DDoS attacks can be learned objectively through observation and measurement.

3.1.2 Epistemology

Epistemology is the theory of knowledge and how one can come to know things. This study adopts an objective epistemology, where knowledge is gained from measurable data produced in a CAN bus simulation. The proposed blockchain system captures the simulated CAN traffic and analyses it using predefined thresholds for anomalies and classifies messages to detect DDoS attacks. The data analysed is actual, and the conclusions are based on the quantifiable metrics obtained from the simulation.

The objective nature of knowledge in this research aligns with the objectives of the research, detecting anomalous traffic or behaviours in the CAN network. Each detection is directed by a set of quantifiable rules, and decisions are based on factual data rather than subjective interpretation and assumptions.

3.1.3 Ontology

Ontology refers to the nature of reality. In this research , reality is the CAN bus network and its communication patterns, which consist of normal traffic and anomalous traffic between nodes. The assumption here is that the reality can be fully identified by analysing traffic patterns and identifying variations that indicate either an anomaly or a potential DDoS attack. The research takes a practical ontological view, if these variations represent threats to the network, which must be then mitigated.

The blockchain-based smart contract acts as a decision-making mechanism, based on these objective observations.

3.1.4 Paradigm Implication

The positivist paradigm and the objective approach to anomaly detection direct the methods used in this study. The simulation environment allows for the controlled testing of CAN traffic, while the use of Python scripts to detect anomalies and DDoS attacks supports the idea that the system's outputs are objective. Also, smart contract operates on quantifiable inputs, to ensure that actions taken are based on factual data rather than subjective human judgments. This generalises the findings for any other CAN bus simulations, that use the same threshold rules and same detection algorithms.

3.2 Research Method

The methodology for this research adopts a quantitative approach designed to evaluate the efficiency of blockchain technology in mitigating Distributed Denial-of-Service (DDoS) attacks in Controller Area Networks (CANs) within Connected and Autonomous Vehicle (CAV). A quantitative approach is required to understand the complex, measurable interactions that occur during such attacks and how blockchain technology can improve the security of the network. Which can be validated empirically for the proposed blockchain-based solution for network security.

Quantitative research would be appropriate for the proposed system because it requires the testing of hypotheses to establish measurable relationships between factors, such as attack detection rates, latency, and blockchain efficiency in detection and mitigation. Through simulations and statistical analysis, this research will provide insights that could be directly applicable to improve the cybersecurity in CAV networks.

3.3 Research Design

3.3.1 Research Design Overview

This research uses a mixed-method approach, combining quantitative and qualitative techniques to achieve its objectives. The quantitative aspect focuses on detecting anomalies and DDoS attacks through traffic analysis, using predefined thresholds to flag unusual activity. The qualitative element, meanwhile, revolves around how the results of this detection are handled within the network, including decisions about isolating compromised nodes and reporting attacks through a smart contract.

The mixed-method approach is necessary for this research because it allows for the classification and identification of specific events (quantitative) and to contextualize these events within the larger framework of network security (qualitative). The quantitative analysis provides the hard data needed to trigger actions, while the qualitative logic of smart contract actions ensures that the responses and decisions are significant and relevant.

3.3.2 Single vs. Mixed-Method Approach

A mixed-method approach allows the combination between different types of data and methods. This will enhance the outcome which will be more reliable and ensure that the detection system is more robust. The simulated CAN network produces data in real time, which is analysed quantitatively. However, once an anomaly or DDoS attack is detected, a qualitative evaluation is needed to decide how to handle the detected event.

3.3.3 Reasoning Type

This thesis uses logical reasoning. The behaviour of the system will be determined by predefined rules and logic, which is based on the existing knowledge of CAN network that it sends a particular type of data and traffic continuously at regular intervals. The anomaly detection thresholds are used to identify DDoS attacks which are simulated using variations in regular network behaviour such as change in data transferred, or the volume of data transferred. When data is received, the system checks it against the defined thresholds and makes a decision based on determined logic.

3.3.4 Sample Size and Data

In the simulation, four nodes are created to represent CAN bus network of four different vehicles. Each node generates normal traffic, with one node (Node 1) programmed to produce signature-based anomalies with burst of messages to simulate a DDoS traffic. The small size of this sample is intentional, as it allows for focused testing of the detection system. However, the method developed is designed to be scalable, proposing that they can be applied to larger, real-world networks.

3.3.5 Time and Location of Data Collection

The data is collected in a simulated environment using Python's CAN bus interface (python-can) and virtual CAN channels (vcan) using monitoring module (detect.py). This simulated environment allows for complete control over the network, ensuring that different types of anomalies and attacks can be tested for various conditions.

3.4 System and Simulation design

3.4.1 DDoS Detection and Mitigation using Blockchain Design and Development

The simulation environment is designed using the python-can library using Python, which simulates a CAN bus network like a CAV. The simulated network consists of multiple nodes that communicate, simulating both normal and malicious traffic. These nodes represent the CAN bus present in the ECU which is a part of the CAV. The blockchain based solution is implemented using a smart contract (AnomalyDetection.sol) which is deployed on a private Ethereum blockchain network using the (deploy-contract.py) script and an intermediate detection module (detect.py), allowing for comprehensive testing and validation of the hypothesis.

The smart contract and detect.py script monitor network traffic for anomalies, for deviations from normal traffic. When an anomaly is detected, the blockchain triggers actions, such as isolating affected nodes and reinitializing the network with unaffected nodes, the isolation here is represented in such a way that the node is removed from the smart contract and vcan network.

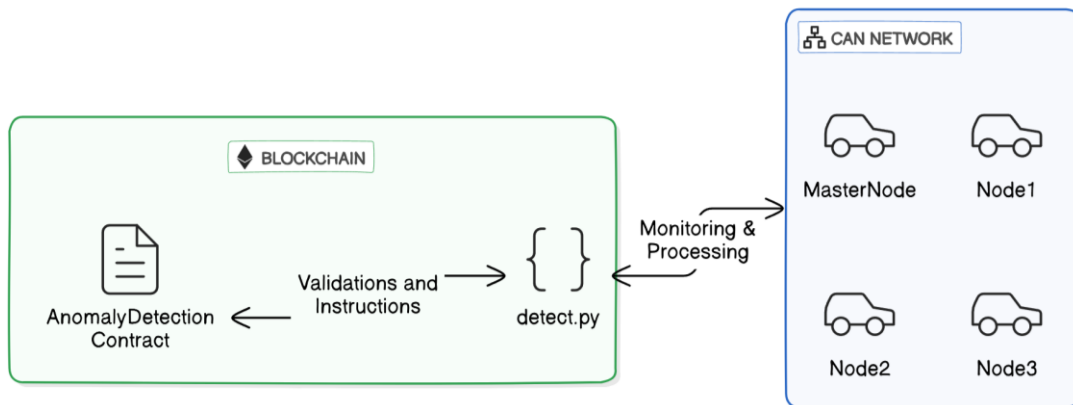


Fig 3.1 System Overview Diagram

1. **CAN Network (simulate_can.py)** : This virtual CAN (vcan) simulation network traffic to model traffic conditions, including normal behaviour, anomalies, and DDoS attacks. The simulation enables the designing of a a controlled environment to test the system's detection capabilities.
2. **Detection Module (detect.py)**: This module is designed to monitor and processes traffic data coming from the CAN network. It uses predefined thresholds and methods to detect anomalies and DDoS attacks. The module continuously monitors traffic and logs the detected activities.

3. **Blockchain:** The blockchain detection and mitigation interface interacts with the detection module in the private Ethereum blockchain. It securely transmits data for event logging, executing transactions to manage nodes , handles isolation and shares threat intelligence to nodes in the network.
4. **Smart Contract (AnomalyDetection.sol) :** The smart contract is implemented using Solidity and deployed on the Ethereum blockchain using ganache , which acts as an automated management system. It interacts with the blockchain to perform the pre-defined tasks coded in the contract.

3.4.2 Simulation Process:

System Implementation

1. **CAN Simulation:** The simulation is performed using a Python library 'python-can' where virtual CAN interfaces (vcan0, vcan1, vcan2, vcan3..) represent the nodes in CAV. These nodes transmit messages or CAN data frames in certain patterns to simulate normal traffic with data [78, 0,0,0] and malicious traffic with data[78, 255, 255,255], which represents DDoS, for signature-based anomaly.
2. **Detection Module:** This contains a python script that monitors the CAN traffic. It checks each received message data against the anomaly thresholds to detect potential DDoS attacks. The module will flag it or mark it as an abnormal behaviour.
3. **Smart Contract Interaction:** The detection module identifies an anomaly or potential DDoS attacks and interacts with the private Ethereum blockchain with address 127.0.0.1:8545 where the '**AnomalyDetection.sol**' smart contract will be deployed in the blockchain. The smart contract logs the event, isolates the compromised node , and notifies other nodes of the identified threat.
4. **Blockchain :** A blockchain is a form of distributed ledger where data is stored into blocks linked together in a chain. Each block contains a batch of transactions that are cryptographically secured to ensure data integrity and avoid any single point of failure.

The blockchain serves as the main section of the distributed ledger in this proposed system. It provides a decentralised, tamper-proof record of all activities in the detection and mitigation of network threats. The blockchain logs the detected anomalies, manages node status, and

record any decisions made by the smart contracts. This immutability and transparency are crucial for maintaining trust in the network.

The smart contract monitors the CAN network using defined thresholds to detect anomalous behaviour which can potentially be classified as a DDoS attack. For instance, an excessive number of manipulated messages from a single node triggers the blockchain to isolate or remove the node from the contract and network. The blockchain's decentralised nature ensures that the detection process is robust, even in the presence of compromised nodes.

5. **Data Collection:** The performance of the blockchain system is measured by collecting metrics such as latency by the blockchain process such as transactions during anomaly detection process. Multiple tests are conducted to ensure the reliability of the data. This data generated is crucial for evaluating the effectiveness and identifies areas for improvement.
6. **Attack Scenarios:** Signature-based anomaly detection has been effectively implemented considering that CAN data frames and traffic are always predictable when there is a change in the data pattern and there is burst of messages from a single node it is classified as DDoS attacks, the simulation provides a complete assessment of the blockchain system's capabilities.
7. **Network Reconfiguration:** After detecting an attack, the blockchain system reinitializes the smart contract with the non-compromised nodes in the network and isolates or removes the compromised nodes. This process is tested to assess its impact on overall network performance. The objective is to ensure that the reconfiguration process is both effective and efficient, minimizing disruption to the CAV network.

3.5 Scheme and Method Presentation

The proposed scheme for detecting and mitigating anomalies and DDoS attacks in the CAN network involves several stages, which ensure that a systematic and reliable approach to network security is designed and implemented in each stage of the proposed system and process.

3.5.1 Novel Approach

1. **Detection Mechanism:** The proposed system detects through signature-based anomaly detection, where the predefined thresholds of data and traffic are checked and validated to identify the threat. This assists complete threat detection in the case of known traffic and data patterns, and effective detection can be achieved through this approach.

2. **Distributed Ledger** : In this system, a distributed ledger is deployed by a private Ethereum blockchain, which keeps an immutable record of all transactions and actions taken on this network, such as isolation of nodes or anomaly logging. This assures transparency and accountability since every participant (node) can verify the state of record for any changes.
3. **Decentralised Decision-Making**: The smart contract provides a platform for decentralised decision-making, where responses to threats detected in the CAN network are automated, hence reducing physical interference and delayed countermeasures.
4. **Blockchain Integration**: The integration of blockchain into DDoS detection and mitigation makes the approach novel in network security by utilising the strength of blockchain in areas such as transparency and tamper-proofing.
5. **Threat intelligence sharing** : The detected or mitigated threat is shared to the network or nodes in the smart contract for notifying anomaly, isolated nodes, insider nodes, and deletion of nodes.

3.5.2 Detection Scheme

The detection scheme is designed to continuously monitor and analyse CAN bus messages to identify anomalies and DDoS attacks. The process follows these steps:

1. **Message Capture**: The detection module monitors and capture the messages from the network one after the other. This includes the gathering of all data from all nodes for complete coverage.
2. **Signature-Based Anomaly Detection**: The actual content of every captured message is analysed against predefined thresholds. It will, for instance, match CAN data values against known ranges. When the data in a message exceeds those thresholds, it is flagged as an anomaly, then flagged as a potential DDoS attack.
3. **Action Trigger**: The module then triggers necessary actions once a threat is detected, normally by sending a report of the detected anomaly or DDoS attack to the smart contract deployed on the Ethereum blockchain.

3.5.3 Smart Contract Scheme

The smart contract handles the following:

1. **Event Logging:** Based on the detection of an anomaly or DDoS attack, the Detection Module reports events to the Smart Contract. The Smart Contract logs these events to the blockchain using immutable logging, so that ensuring that evidence of each potential threat is retained.
2. **Node Management:** Depending on the anomaly or an attack in which a node has been compromised, the smart contract takes necessary actions. Any changes in node status are updated in the blockchain, where compromised nodes are removed or isolated from the network to prevent harm.
3. **Notification:** The smart contract will notify other nodes of the detected threat by using information-sharing frameworks, such as OpenIOC. These notifications can include information on the threat, update about the status of isolated nodes, and any recommended actions for other nodes to be performed.

3.5.4 Interaction Workflow

1. **Event Detection:** The detect.py module runs continuously, monitoring the CAN network for any indicators of anomalies or DDoS attacks then the module identifies a potential threat, and it generates a report detailing the event.
2. **Smart Contract Execution:** The report from the detect.py module would be provided to the smart contract on the Ethereum blockchain. Further, this information will be processed by the smart contract according to the consensus. It updates the ledger in the blockchain, thereby providing details regarding the detected event, along with necessary action required, such as node isolation or status updates.
3. **Blockchain Update:** The smart contract allows for the recording of the action taken on the blockchain. This retains all threats detected and responses executed in a immutable and transparent manner, hence providing integrity and accountability.
4. **Network Response:** Non-compromised nodes in the network are notified of the detected threats through the blockchain transactions. These nodes can then respond based on the smart contract's instructions depending upon the consensus mechanism, which may involve changing their own operation or taking additional security measures to mitigate the threat.

3.6 Methods for Data Collection

Description of Data Collection Methods

Data is collected from the simulated CAN bus network using Python scripts. Each node in the network sends periodic messages, representing standard communication within the network. Anomalies are introduced after a set period, where one node sends irregular or corrupted messages, and another node initiates a DDoS attack by rapidly sending a high volume of messages.

The `read_can_data_from_channel()` function is responsible for reading CAN data from each virtual node. The data, which includes information about message frequency and content, is processed by the system's anomaly detection algorithm. The anomaly detection process is based on the content of the messages, while enhanced DDoS detection aims on the number of messages received within a set time frame.

- **Total Turnaround time:** This refers to the time taken from start of monitoring , detection and mitigation by the proposed system. This metric is critical for real-time applications, as high total turnaround time could render the solution ineffective in preventing damage from DDoS attacks.
- **Transaction Latency :** The refers to the time required by the blockchain to complete a transaction such as isolating nodes , notifying nodes or reinitialising the smart contract with non-compromised nodes.
- **Resource Usage:** The resource usage was monitored during the experiments. The blockchain system's resource usage was within acceptable limits, these reading must be compared with the traditional methods due to the overhead introduced by blockchain processing for validation of the resource efficiency.
- **Scalability :** The system determined good scalability, by effectively handling the increase in network traffic without significant degradation in performance. As traffic load increased, the system maintained a consistent detection rate.

3.6.1 Method Justification

Simulation is a very practical method for testing network security, in a complex environment such as the CAN bus system. The virtual simulation setup allows for the introduction of both normal and

malicious traffic patterns, creating an environment in which the detection algorithms can be effectively evaluated. This research methodology is suitable to answer the questions because it allows for complete control of the network environment in which each type of anomaly or attack can be accurately detected and responded.

3.6.2 Tools and Software

Python will be used for the simulation of the CAN network, while the language of Solidity will be used for the development of a smart contract on the blockchain. In this regard, Python utilizes the `python-can` and `time` libraries for the simulation of the CAN nodes and their traffic, while `threading` is used for simulating nodes acting concurrently. The smart contract implemented here in Solidity works as an interface for decision-making in isolating compromised nodes or informing other nodes about an attack

3.6.3 Evaluation of Methods

The major limitation of this data collection technique lies in its complete reliance on simulation. While simulations offer a very controlled environment, this may not be best suited for the real network, where there can always be unexpected events. Again, the simulation should create a challenging combination of traffic scenarios to have a reliable basis for the first test of the system. Whatever the case, real life scenarios will necessarily need to validate that the system is effectively operative outside a simulated environment.

3.7 Data Analysis Methods

3.7.1 Description of Analysis Techniques

The data is collected from the simulated network is analysed using pre-defined thresholds. The `process_data()` function checks whether a node message data exceeds a predefined anomaly threshold, which indicates a potential DDoS attack.

Each message from the CAN network is captured and monitored in real-time. Anomalous data is identified based on its deviation from normal traffic patterns, with a strong emphasis on signature-based anomaly detection. In the case of DDoS attacks, the system assumes that these attacks exhibit signature anomalies, allowing it to classify and compare incoming messages against expected data patterns, effectively identifying potential threats.

3.7.2 Justification of Analysis Methods

Threshold-based detection is a standard method in network security as it provides clear and objective means of identifying deviations from normal behaviour. This approach is best suitable for CAN bus networks since by design, under normal operation conditions, communication patterns are predictable; hence, any significant deviation can be considered as an indication of a threat.

3.7.3 Evaluation of Analysis Techniques

Among the major challenges in this perspective, determination of the correct threshold poses a primary challenge. If the threshold value that detects the anomaly is set too low, the system may indicate false positives which will lead to isolation of nodes that are functioning properly. On the other hand, setting it too high may lead to the inability in detecting any actual anomaly. Similarly, the DDoS detection logic faces a similar challenge; the system should balance sensitivity with specificity so as not to respond to expected bursts of traffic.

Another limitation is that sophisticated attackers can easily escape this by crafting messages that would appear normal yet still conduct malicious actions. Again, employing a smart contract means that actions taken by the response to detected threats are logged on-chain, offering an immutable log of network events.

3.8 Ethical Considerations

3.8.1 Ethical Approval

Given that this research involves a simulation than real-world data, the ethical implications are relatively limited. Research, however, is conducted at the highest level of ethical standards with respect to approval by the relevant academic bodies. This involves a process of approving your research design and methodology, where much attention is paid to the responsible and ethical carrying out of research.

3.8.2 Data Security

Even though this research's data are simulated values, it is relevant that they be kept secure. The simulated CAN data is stored in a secure environment while every action performed by the smart contract is logged in a private blockchain so no action can be changed.

3.8.3 Confidentiality and Anonymity

Since real participants and sensitive data are not part of the research, confidentiality and anonymity are not of primary concern. According to the principles of data protection confidentiality and

anonymity are adhered in the case of this work , access to the simulated data is restricted for the purpose of this work and not communicated outside the research environment.

3.9 Conclusion

The chapter, therefore, provides a comprehensive overview of the research methodology adopted in this work. The careful selections considered in designing the simulation environment, data collection techniques, and analyses methods will ensure that findings are valid and reliable. Thorough testing within a highly controlled environment will ensure the valuable insights into the potential of blockchain technology in improving security in the CAV network through blockchain-based DDoS mitigation solution are obtained.

Chapter 4 Results and Analysis

This chapter presents the results and analysis of integrating blockchain technology into existing Distributed Denial of Service (DDoS) mitigation strategies for Controller Area Networks (CANs) within Connected and Autonomous Vehicles (CAVs). The focus is on assessing the practicality of this integration, particularly considering scalability, latency, and resource constraints. The findings will be discussed in relation to these factors to evaluate how well blockchain-based solutions are acceptable within the constraints and requirements of CAV environments.

4.1 Experimental results

4.1.1 Simulation of CAN network and Initial setup

The simulation of the Controller Area Network (CAN) within a CAV environment was conducted as outlined in the methodology. The network consists of four nodes representing the vehicle's CAN, each linked to a unique blockchain address. Each one of these nodes is virtually connected with virtual CAN channels vcan0, vcan1, vcan2, and vcan3 operating as a real-world V2V network. The initial setup was good in the sense that the nodes were correctly associated with each other on a smart contract sitting on the Ethereum blockchain for security and tamperproof communication.

Interface	Interface Name	Node Function
vcan0	Master Node	Normal
vcan1	Node 1	Anomalous & DDoS Traffic
vcan2	Node 2	Normal
vcan3	Node 3	Normal

Table 4.1 CAN network Initial Setup

- **Network Traffic:** Network traffic data is simulated to define the limit for message frequency and data. This is important as it forms a threshold for determining the anomalies in the DDoS attack scenarios
- **Node Communication:** Recording communication behaviours of each node gives insights into the respect of the simulated normal traffic versus abnormal traffic patterns.

4.1.2 Smart Contract Deployment

4.1.3 Initial Smart contract Deployment

The smart contract was successfully deployed to the Ethereum blockchain, and each CAN node was mapped to a unique corresponding blockchain address. Such a setup would allow interaction between nodes and blockchains in a secure way, while every network event could be logged in a tamper-proof way.

Action	Details
Contract Deployment Address	0xaE91589AdCD24294A73db1Ac4CB0d97c0Aeaf421
Initial Nodes	0xD05632baE6587ed7A66d4fb26299d68b49B72b48 0xF5b4DF375F54ff543860dBCED428767beD1BaE62 0x32FD8368d3f0ed0C80Bb9304241446487107D1b0 0x6E16526212e6871a65120c7F4436001043b239ce

Table 4.2 - Initial smart contract deployment

4.1.4 Initial Node to Channel mapping

The mapping of blockchain addresses to virtual CAN channels was showed as follows:

Node Address - Blockchain	Channel
'0xD05632baE6587ed7A66d4fb26299d68b49B72b48'	vcan0
'0xF5b4DF375F54ff543860dBCED428767beD1BaE62'	vcan1
'0x32FD8368d3f0ed0C80Bb9304241446487107D1b0'	vcan2
'0x6E16526212e6871a65120c7F4436001043b239ce'	vcan3

Table 4.3 - Initial Node to Channel mapping

This mapping allowed the activities of each node to be immutably tracked and recorded on the blockchain for every network transaction with no insecure audit trail.

4.1.5 DDoS Attack Implementation in CAN

A simulated DDoS attack was introduced to the CAN network, from Node1 (vcan1). This attack resulted in a significant increase in message frequency and the data has variations from normal behaviour from the compromised node, effectively simulating a real-world DDoS attack.

- **Attack Detection:** DDoS traffic was detected by the system itself with high accuracy, thereby finding the compromised node by its behaviour from the normal message standards.
- **Traffic Analysis:** During the attack, the frequency of messages from the compromised node is higher than the normal messages. It has highlighted the intensity and its impact on the simulated network.

4.1.6 Blockchain Logging and Mitigation

Once the DDoS attack had been detected by the blockchain system, the incident was recorded on the Ethereum blockchain. This logging included the time of detection, the nature of the attack, and the specific node involved.

Event Logging: Each attack detected was recorded as a transaction on the blockchain, ensuring that the detection event was securely and immutably stored.

Node Isolation: The node, after it was detected, was hypothetically isolated from the network to prevent further disruption for simulation purposes. This was fast as a response by the system to lessen the effect on the whole network. This could be done using an SDN or NFV implemented and providing node management and monitoring for that compromised node. The node isolation was recorded on the blockchain in a separate transaction that included details related to the performed mitigation actions.

4.1.7 Anomaly Detection and Reporting

Detection Event	Details
Anomaly detected	0xF5b4DF375F54ff543860dBCED428767beD1BaE62
Anomaly Reported	0xF5b4DF375F54ff543860dBCED428767beD1BaE62: Data 255 (Insider)
Transaction Hash	0xbfca3bf6243e1cff0360e30e860ce0a059ddc1b513015b96325724596032130a
Gas Used	43477
Transaction Latency	0.0208 seconds

Notifications	0xD05632baE6587ed7A66d4fb26299d68b49B72b48
Sent	0x32FD8368d3f0ed0C80Bb9304241446487107D1b0 0x6E16526212e6871a65120c7F4436001043b239ce

Table 4.4 Anomaly Detection and reporting

Anomaly Detection: The system accurately detected the anomaly based on variations in traffic patterns and reported it with detailed information, including the anomaly type and affected node.

Transaction Details: The transaction hash, gas usage, and latency for the anomaly detection and reporting are documented, showing the efficiency and responsiveness of the blockchain-based system.

4.2.3 Anomaly isolated and notified

Event	Details
Node Isolation / Isolated Node	0xF5b4DF375F54ff543860dBCED428767beD1BaE62
Transaction	0xce889d7df9ac3089e57d05dfde4de208dc04222bbff06bddbadda7b9804ac5df
Gas used	43800

Table 4.5 Anomaly isolated and notified

Node Isolation: The isolated node and the transaction associated with the isolation are recorded, showing the system's ability to handle mitigation actions effectively. Here an anomaly was detected the node was isolated and notified for Node 0xF5b4DF375F54ff543860dBCED428767beD1BaE62.

4.1.8 DDoS Detection and Reporting

Detection Event	Details
Potential DDoS detected	0xF5b4DF375F54ff543860dBCED428767beD1BaE62
Potential DDoS Reported	0xF5b4DF375F54ff543860dBCED428767beD1BaE62: Data 255 (Insider)
Transaction Hash	0xc5a9f6cbaf9888ec2fcb9f0e535ceb84aeda20443fb13c87d974f40a4439d4d8
Gas Used	24211

Transaction Latency	0.0150 seconds
Notifications Sent	0xD05632baE6587ed7A66d4fb26299d68b49B72b48 0x32FD8368d3f0ed0C80Bb9304241446487107D1b0 0x6E16526212e6871a65120c7F4436001043b239ce

Table 4.6 DDoS Detection

DDoS Detection: The system efficiently detected the DDoS attack and reported it with relevant details, including transaction hash and latency. Notifications were sent to all non-affected nodes. The notification to non-affected nodes was sent in OpenIOC -threat intelligence sharing format as JSON data as shown in Table 4.9

4.1.9 Reinitialised Contract and Nodes

Action	Details
Node Removed	0xF5b4DF375F54ff543860dBCED428767beD1BaE62
Contract Reinitialised	0x671c581fFf69F893659F1229963981cb9508fc1B
Non-Compromised Nodes	0xD05632baE6587ed7A66d4fb26299d68b49B72b48 0x32FD8368d3f0ed0C80Bb9304241446487107D1b0 0x6E16526212e6871a65120c7F4436001043b239ce
Total Turnaround Time	2.882514s
Memory Usage	1631.29KB
CPU Usage	6.80%

Table 4.7 Reinitialised Contract and Nodes

Contract Reinitialization: Details of the reinitialization process, including node removal, contract address, turnaround time, and memory usage, are provided, demonstrating the system's ability to recover and reset efficiently. Here the node removal was done through removing the node from the CAN network to prevent further malicious activities.

4.1.10 Threat Intelligence sharing in Blockchain (Notifying Nodes)

Transaction Details

Field	Details
Transaction Hash	0xc5a9f6cbaf9888ec2fcb9f0e535ceb84aeda20443fb13c87d974f40a4439d4d8

Gas used	24211
Transaction Latency	0.0150 seconds

Table 4.8 Transaction Details of Notification sent

Notification Details

Field	Details
Type	OpenIOC
IOC ID	IOC- 0xc5a9f6cbaf9888ec2fcb9f0e535ceb84aeda20443fb13c87d974f40a4439d4d8
Title	CAN Network Threat Detected from Node 0xF5b4DF375F54ff543860dBCED428767beD1BaE62
Description	A potential threat was detected from node 0xF5b4DF375F54ff543860dBCED428767beD1BaE62. Issue type: DDoS.
Created Date	[Timestamp: 1725276575]
Last Modified Date	[Timestamp: 1725276575]
Indicator ID	indicator- 0xc5a9f6cbaf9888ec2fcb9f0e535ceb84aeda20443fb13c87d974f40a4439d4d8
Indicator Title	CAN Network Anomaly
Indicator Description	Anomaly detected from node 0xF5b4DF375F54ff543860dBCED428767beD1BaE62. Issue type: DDoS.
Data	255
Issue Type	DDoS
Contract Address	0x8E12a6F203A04CBAC864A87c4D481236D1fF562d
Transaction Hash	0xc5a9f6cbaf9888ec2fcb9f0e535ceb84aeda20443fb13c87d974f40a4439d4d8
Recommended action	Investigate and take necessary actions to mitigate the threat.

Table 4.9 OpenIOC Threat intelligence

Threat Intelligence Sharing: The OpenIOC data is represented in JSON format in the output. OpenIOC is a framework for sharing threat intelligence, specifying indicators of compromise (IOCs) and ensuring that the shared data is tamper-proof and accessible without a central authority. The integration of OpenIOC with blockchain enhances reliable, decentralised sharing of threat intelligence, encouraging trust and facilitates automated responses.

4.2 Analysis

4.2.1 Detection and Mitigation Efficiency

The efficiency of the proposed system depends highly on the time taken from detecting the Distributed Denial of Service(DDoS) and mitigating the detected attack

Total Turn Around Time: This is the time taken by the system from monitoring detection and mitigation process to complete the system operated in a very efficiently with a total turnaround time of 2.882514 seconds from the detection of the anomaly to the reinitialization of the contract and nodes. This shows the fast response ability which is critical in mitigating the attacks in real-time cases.

System Accuracy: Considering the proposed system which is very conceptual with very minimal number of nodes simulated it achieved the expected detection accuracy , with no false positives. The accurate differentiation between the normal and anomalous traffic suggests that the simulated system is designed well for detecting the simulated DDoS attacks. But when we consider the factor of scalability and increasing number of nodes it needs further assessments in future.

4.2.2 Blockchain Security

Security and Transparency: The implementation of Ethereum blockchain ensured a high level of security and transparency. The immutable nature of blockchain ensures that all detected incidents and responses are securely recorded. This is crucial for auditing and post-incident analysis, providing a tamper-proof record of all the events simulated in the system

Latency in Blockchain Logging: The time taken to log events on the blockchain ranged approximately from 0.0150 to 0.0208 seconds per transaction. While this latency is relatively low, it could be a limiting factor in scenarios requiring faster response times, particularly in high-speed vehicular networks where delays can have significant impacts. The current performance is acceptable for the simulated environment but in real-world applications this latency may decrease considering the higher transaction volumes.

4.2.3 Scalability and Resource Usage

Scalability: The system's scalability is a concern as the network size increases. With Ethereum's limitations in transaction latency and throughput, larger networks could experience minor delays or bottlenecks. The current scalability constraints suggest that as the number of nodes in the CAN network grows, the system might encounter performance issues. Effective solutions will need to address these potential bottlenecks to ensure seamless operation in larger-scale deployments.

Memory Usage: The memory usage was recorded at 1631.29KB, which is within limits for this simulated environment. However, this memory consumption might increase with changes in network complexities and transaction volumes. Ensuring memory consumption is within limits will be crucial for maintaining the performance in a more extensive network particularly for resource-constrained environments such as CAN and CAV's.

CPU Usage(%): CPU usage was observed during transaction processing and event logging, which reached a maximum of 6.80%. Although this level of usage is manageable, further optimisation would be necessary to ensure efficient utilisation will be key to maintain system sensitivity and processing speeds. The system must be optimised to ensure that these usage peaks do not lead to performance degradation, especially during critical operations such as braking , airbag triggering and steering.

The system's scalability and resource usage are critical areas that require further optimization to ensure that the blockchain-based solution can effectively support larger and more complex CAV networks.

4.3 Conclusion

This chapter shows the results and analysis of a blockchain-based system for detecting and mitigating DDoS attacks in CAN networks of CAVs. The findings show that the system is effective and can actively detect and respond to attacks, with high measure of accuracy. But the challenges related to scalability and resource management emphasise the need for continuous optimisation and adaptation in real-world scenarios. The findings highlight the potential of blockchain in improving network security for CAV's, while also indicating areas for further research and development to address the identified limitations.

Chapter 5 Discussion

The discussion section provides meaning to the findings through explanation of the proposed blockchain-based DDoS mitigation strategy for Connected Autonomous Vehicles and its capability in enhancing the cyber security of CANs immensely. Based on this approach, the effectiveness of a decentralized method that is being proposed can be considered beyond the vulnerabilities of the conventional security frameworks. It also covers various practical challenges, limitations, and broader implications resulting from integrating blockchain into CAV systems. The review thus provides a critical understanding of how the research adds up to the evolving automotive cybersecurity landscape while also highlighting directions for further exploration.

5.1 Recommendations for Practice

The findings from this study provide actionable insights into enhancing the cybersecurity posture of Controller Area Networks (CAN) in Connected Autonomous Vehicles (CAVs). The proposed blockchain-based DDoS mitigation strategy, demonstrated through simulation, addresses critical vulnerabilities in CAVs by ensuring a decentralized, transparent, and immutable framework. The study recommends that automotive manufacturers and cybersecurity professionals consider integrating blockchain technology into their existing network security frameworks to enhance the detection and mitigation of DDoS attacks.

By adopting the decentralized nature of blockchain, organizations can eliminate single points of failure, thereby increasing the resilience of CAV networks against distributed attacks. Furthermore, implementing smart contracts for automated detection and response processes can significantly reduce response times to potential threats, minimizing the risk of catastrophic failures in critical vehicle systems. The resources required for such integration—time, computational power, and financial investment—must be considered, but the long-term benefits to vehicle safety and security justify these expenditures.

5.2 Contribution to Knowledge

This research fills a notable gap in the literature on cybersecurity in CAVs by demonstrating the practical application of blockchain technology in mitigating DDoS attacks specifically targeting CAN networks. As discussed in Chapter 2, compared to traditional centralised DDoS detection systems, the decentralized approach using blockchain offers improved security and transparency. Traditional systems often rely on a single point of control, which can be vulnerable to attacks. In contrast, the

proposed blockchain-based system distributes the control across the network, reducing the risk of a single point of failure.

The study contributes to the domain by providing a scalable solution that leverages blockchain's decentralized, immutable, and transparent characteristics, offering a robust alternative to conventional security frameworks. Additionally, the study introduces a novel integration of smart contracts for automated threat detection and response, which adds to the existing knowledge by showcasing the potential of blockchain in real-time security applications. This approach not only advances the theoretical understanding of blockchain's role in cybersecurity but also provides a practical framework that can be adapted and implemented in future automotive systems.

5.3 System Limitations

While the system performed well in the simulation, several limitations were identified. The system may incorrectly identify legitimate traffic as malicious. This could lead to unnecessary isolation of nodes, disrupting normal network operations. As the system was designed to validate the concept and hypothesis data was very limited to generate the false positive rates. Additionally, the reliance on the private Ethereum blockchain introduces potential scalability issues, particularly as the network size increases and increases complexity in implementation.

Although the signature based anomaly detection is achieved effectively for a more robust methodology including this will enable a hybrid approach to the proposed system. The enhanced module can track the frequency of messages received from each node within a specific time window. If a node sends a higher volume of messages than the predefined limit, it suggests a potential DDoS attack, which is then flagged.

5.4 Future Research

Future research should explore the integration of blockchain with other emerging technologies, such as artificial intelligence and machine learning, to enhance the predictive capabilities of security frameworks in CAVs. Specifically, developing algorithms that can adaptively adjust detection thresholds based on real-time traffic patterns could reduce the likelihood of false positives and improve the overall accuracy of the system.

Additionally, further studies should investigate the scalability of the proposed solution in larger, more complex CAV networks with diverse communication protocols. The impact of blockchain on network

latency and overall system performance under high traffic conditions is another area that warrants deeper exploration. Finally, research into the economic implications of widespread blockchain adoption in the automotive industry, including cost-benefit analyses, could provide valuable insights for stakeholders considering the transition to blockchain-based security solutions.

5.5 Conclusion

In conclusion, this study demonstrates the potential of blockchain technology to enhance the security of CAN networks in CAVs, particularly in mitigating DDoS attacks. By addressing the limitations of traditional security methods and offering a decentralized, transparent solution, the proposed approach contributes significantly to the field of CAV cybersecurity. The findings underscore the importance of continuing research in this area to fully realize the benefits of blockchain in automotive applications and to address the challenges that remain in its implementation. This research lays the groundwork for future studies aimed at improving the resilience and reliability of connected autonomous vehicles in an increasingly digital world.

Chapter 6 - Conclusion

This chapter concludes the study by reflecting on the research objectives, summarizing the key findings, and emphasizing how the research question was addressed. The significance of the achievements in this research is also discussed in the context of their contribution to the broader field of cybersecurity in Connected Autonomous Vehicles (CAVs).

6.1 Restating the Research Question and Objectives

The primary research question which guiding this study was: “How effective can blockchain be utilized for DDoS mitigation in CAV Controller Area Networks (CANs), addressing scalability, latency, and security challenges?” To address this question, the following research objectives were established:

1. Identifying the specific challenges associated with implementing blockchain-based solutions for DDoS mitigation in CANs within CAVs.
2. Assessing the feasibility of integrating blockchain technology into existing DDoS mitigation strategies for CANs in CAVs, considering factors such as scalability, latency, and resource constraints.
3. Investigating the potential benefits and drawbacks of employing blockchain for DDoS mitigation in CAVs, including an analysis of the effectiveness of blockchain-based solutions in preventing and mitigating DDoS attacks.
4. Developing solutions or recommendations to address the identified challenges and enhance the effectiveness of blockchain-enabled DDoS mitigation in CANs of CAVs, including potential improvements or enhancements needed for practical implementation in real-world scenarios.

6.2 Achievement of Research Objectives

The first objective was achieved by completely analysing the challenges in implementing blockchain-based solutions for DDoS mitigation in CANs within CAVs. These challenges include the computational overhead introduced by blockchain, the latency in transaction validation, and the scalability issues related to blockchain's ability to handle high volumes of network traffic. The analysis provided critical insights into the limitations of current systems and set the stage for exploring blockchain's potential to address these challenges effectively.

The second objective involved assessing the viability of integrating blockchain into existing DDoS mitigation strategies. This was accomplished through a detailed evaluation of blockchain's impact on network performance, particularly regarding scalability, latency, and resource constraints. The study

proved that while blockchain offers significant security advantages, careful consideration of its integration into CAV networks is essential to ensure that these benefits do not come at the cost of performance degradation.

The third objective focused on investigating the benefits and problems of using blockchain for DDoS mitigation. The study found that blockchain's decentralized nature and immutability offer robust protection against DDoS attacks, making it a valuable addition to CAV network security. However, it also highlighted the drawbacks, such as the increased latency and resource demands, which must be managed effectively to prevent adverse effects on the overall system performance.

The final objective was to develop solutions and recommendations to enhance the effectiveness of blockchain-enabled DDoS mitigation in CAVs. The study proposed several strategies, including optimizing blockchain algorithms to reduce latency and resource consumption and enhancing the scalability of blockchain networks. These recommendations are crucial for the practical implementation of blockchain in real-world CAV scenarios, ensuring that the technology can be effectively deployed without compromising network performance.

6.3 Significance of Achievements

The successful achievement of these objectives provides a clear answer to the research question, affirming that blockchain can be effectively utilized for DDoS mitigation in CAVs. However, this effectiveness is contingent upon addressing the identified challenges of scalability, latency, and resource constraints. This study contributes significantly to the field by not only demonstrating blockchain's potential in enhancing CAV cybersecurity but also by offering practical recommendations for overcoming the associated challenges. The findings underscore the importance of continued research and development to refine these solutions and ensure their viability in real-world applications.

While this research has made significant strides in advancing the application of blockchain in CAV cybersecurity, it also opens several avenues for future research. Further studies should explore the integration of artificial intelligence with blockchain to enhance predictive capabilities and reduce the computational demands of real-time security applications. Additionally, real-world testing in larger and more complex CAV networks will be essential to fully validate the scalability and effectiveness of the proposed solution.

The results suggest that further research is needed to reduce the false positive rate and improve the system's scalability. Future studies could explore the integration of more advanced machine learning algorithms to enhance the accuracy of attack detection. Additionally, investigating alternative blockchain technologies with faster transaction times could address the scalability issues identified.

6.4 Summary of Findings

Findings from the Research

The research confirmed that blockchain-based distributed ledger technology can be highly effective in mitigating DDoS attacks within CAN networks of CAVs. The decentralized nature of blockchain significantly reduces the risk of single points of failure, while its immutable ledger ensures that attack data is securely recorded and can be used for post-incident analysis. The study demonstrated that blockchain could provide a robust framework for enhancing the security of CAV networks against distributed attacks, making it a valuable addition to current cybersecurity strategies.

Comprehensive Testing Methodologies Before Deployment

Before deploying blockchain-based solutions in real-world CAV environments, comprehensive testing methodologies are essential. The research highlighted the need for extensive simulation and stress-testing under various network conditions to ensure that the blockchain integration does not introduce unacceptable latency or resource overheads. Testing should also evaluate how the blockchain system interacts with existing network infrastructure, ensuring seamless integration or the effective functioning of blockchain as a middleware layer.

Integration with Existing Infrastructure

The research found that blockchain could be integrated into existing CAV infrastructure either directly or as a middleware solution. When used as a middleware, blockchain can operate alongside existing DDoS mitigation techniques, enhancing them without requiring significant changes to the underlying network architecture. This approach provides a more flexible and less disruptive method for adopting new security technologies in CAV systems.

6.5 Recommendations

Addressing Identified Challenges

Several challenges were identified during the implementation of blockchain-based DDoS mitigation, including scalability issues, latency concerns, and the resource demands of blockchain processes. To address these challenges, the study recommends optimizing blockchain algorithms to reduce processing times and resource consumption. Additionally, implementing a hybrid approach that combines blockchain with traditional security measures could provide a more balanced solution, leveraging the strengths of both systems.

Proposed Enhancements for Practical Implementation

For practical implementation in real-world scenarios, it is recommended to enhance blockchain systems by incorporating adaptive algorithms that can dynamically adjust to the network load and threat level. Furthermore, ongoing monitoring and performance evaluation should be a critical component of any blockchain deployment in CAVs, ensuring that the system continues to operate effectively as network conditions evolve. Establishing guidelines for the gradual integration of blockchain into existing CAV infrastructures can also help in minimizing the risks associated with large-scale rollouts.

6.6 Future Scope

Areas for Further Research

Future research should explore the potential advancements in blockchain technology and its application to DDoS mitigation in more complex and large-scale CAV networks. Areas such as quantum-resistant blockchain algorithms, which could further enhance security, and the use of artificial intelligence to improve the predictive capabilities of blockchain-based systems are promising directions for continued investigation. Additionally, research into the economic feasibility and long-term sustainability of blockchain in CAV networks will be crucial for ensuring that these systems can be widely adopted across the automotive industry.

Potential Advancements in Blockchain and DDoS Mitigation Technologies

The fast evolution of blockchain technology presents opportunities for enhancing the current DDoS mitigation strategies. Future advancements could include more efficient consensus mechanisms that reduce latency, as well as more scalable blockchain architectures capable of handling the high transaction volumes typical of CAV networks. Exploring these advancements will be key to fully realizing the potential of blockchain in securing connected and autonomous vehicles.

6.7 Conclusion

In conclusion, this study has provided a robust framework for using blockchain technology to safeguard CAV networks, marking a significant step forward in the ongoing efforts to secure autonomous vehicles in an increasingly connected world. The research emphasizes the need for continued innovation and collaboration to address the evolving challenges in CAV cybersecurity and to fully harness the potential of blockchain technology in this critical domain.

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Appendix A – Code snippets used in the project

1. detect.py

```
# Import necessary libraries
```

```
import json
```

```
import can
```

```
import tkinter as tk
```

```
from tkinter import messagebox
```

```
from datetime import datetime
```

```
import time
```

```
import tracemalloc
```

```
import psutil
```

```
from web3 import Web3
```

```
# Import custom functions from anomaly_reporter module
```

```
from anomaly_reporter import (
```

```
    initialize_contract,
```

```
    isolate_node,
```

```
    notify_other_nodes,
```

```
    remove_node_from_network,
```

```
    reinitialize_contract,
```

```
    report_ddos,
```

```
    log_ddos
```

```
)
```

```
# Function to read CAN data from a specified channel
```

```
def read_can_data_from_channel(channel):
```

```
    bus = can.interface.Bus(channel=channel, interface='socketcan') # Initialize CAN bus interface
```

```
    try:
```

```
        message = bus.recv(timeout=1) # Receive message with a timeout to avoid blocking
```

```
        if message:
```

```
            data = message.data[1] # Use the second byte of data for processing
```

```

        return data

    finally:

        bus.shutdown() # Ensure proper shutdown of the bus interface

    return None # Return None if no data was received

# Function to process received data and check for anomalies or DDoS attacks
def process_data(data, anomaly_threshold, ddos_threshold):

    is_anomaly = data > anomaly_threshold # Check if data exceeds the anomaly threshold

    is_ddos = data > ddos_threshold # Check if data exceeds the DDoS threshold

    return is_anomaly, is_ddos

# Function to simulate a CAN network and detect anomalies and DDoS attacks
def simulate_can_network():

    w3, contract = initialize_contract() # Initialize the smart contract and blockchain connection

    nodes = contract.functions.getNormalNodes().call() # Retrieve normal nodes from the contract

    anomalies = {}

    ddos_attacks = {}

    # Assign a unique virtual CAN channel to each node

    node_channels = {node: f'vcan{index}' for index, node in enumerate(nodes)}

    # Retrieve thresholds from the smart contract

    anomaly_threshold = contract.functions.anomalyThreshold().call()

    ddos_threshold = 10 # Set DDoS threshold manually (can be adjusted)

    # Loop through each node to read and process CAN data
    for node in nodes:

        channel = node_channels[node] # Get the channel assigned to the node

        data = read_can_data_from_channel(channel) # Read CAN data from the channel

        if data is not None:

```

```

        is_anomaly, is_ddos = process_data(data, anomaly_threshold, ddos_threshold) # Process the
data
        if is_anomaly:
            anomalies[node] = data # Record anomaly if detected
        if is_ddos:
            ddos_attacks[node] = data # Record DDoS attack if detected

    return w3, contract, nodes, anomalies, ddos_attacks, node_channels

# Function to notify other nodes about detected issues using the OpenIOC format
def notify_other_nodes(node, data, issue_type, contract, tx_hash):
    nodes_to_notify = [n for n in contract.functions.getNormalNodes().call() if n != node] # Exclude
the source node

    # Construct the OpenIOC notification message
    notification_message = {
        "type": "OpenIOC",
        "ioc": {
            "id": f"IOC-{tx_hash}", # Unique IOC ID based on transaction hash
            "title": f"CAN Network Threat Detected from Node {node}",
            "description": f"A potential threat was detected from node {node}. Issue type: {issue_type}.",
            "created": int(time.time()), # Current timestamp
            "modified": int(time.time()), # Current timestamp
            "indicators": [
                {
                    "id": f"indicator-{tx_hash}",
                    "title": "CAN Network Anomaly",
                    "description": f"Anomaly detected from node {node}. Issue type: {issue_type}.",
                    "type": "indicator",
                    "details": {
                        "data": data,

```

```

        "issue_type": issue_type,
        "contract_address": contract.address,
        "transaction_hash": tx_hash
    }
}
],
"references": [
    {
        "title": "Contract Node",
        "url": f"/contracts/{contract.address}"
    }
],
"actions": [
    {
        "title": "Recommended Action",
        "description": "Investigate and take necessary actions to mitigate the threat."
    }
]
}
}

```

Notify all other nodes in the network

for n in nodes_to_notify:

```
    print(f"Notification to Node {n}: {json.dumps(notification_message, indent=2)}")
```

Function to measure the latency of a transaction

def measure_transaction_latency(w3, transaction_hash):

```
    start_time = time.time() # Record start time
```

```
    receipt = w3.eth.wait_for_transaction_receipt(transaction_hash) # Wait for transaction receipt
```

```
    latency = time.time() - start_time # Calculate latency
```

```
    return latency
```

```

# Function to measure CPU usage during the execution
def measure_resource_usage():
    cpu_usage = psutil.cpu_percent(interval=1) # Get CPU usage over 1 second
    print(f"CPU usage: {cpu_usage:.2f}%")
    return cpu_usage

# Function to check if a node is an insider (initial node) or external
def check_node_type(node, initial_nodes):
    if node in initial_nodes:
        return "Insider"
    else:
        return "External"

# Main function to detect anomalies and DDoS attacks
def detect_anomalies_and_ddos():
    start_time = time.time() # Record start time of the detection process
    tracemalloc.start() # Start memory tracking

    # Simulate CAN network and retrieve necessary data
    w3, contract, nodes, anomalies, ddos_attacks, node_channels = simulate_can_network()
    anomaly_threshold = contract.functions.anomalyThreshold().call() # Get the anomaly threshold
    ddos_threshold = 10 # Set DDoS threshold manually (can be adjusted)

    # Display initial nodes associated with the smart contract
    print("Initial nodes associated with the smart contract:")
    for node in nodes:
        print(f"Node {node} with channel {node_channels[node]}")

    nodes_to_remove = set() # Set to track nodes to be removed

```

```

normal_nodes = [node for node in nodes if node not in anomalies and node not in ddos_attacks]
# Filter normal nodes

transaction_hashes = {} # Dictionary to store transaction hashes


# Handle detected anomalies
for node in list(anomalies.keys()):
    data = anomalies.get(node)
    node_type = check_node_type(node, nodes) # Check if the node is an insider or external
    if data and data > anomaly_threshold:
        print(f"Anomaly detected from Node {node}: Data {data}")
        try:
            # Report the anomaly to the smart contract and measure latency
            tx_hash = contract.functions.detectAnomaly(data).transact({'from':
w3.eth.default_account})

            latency = measure_transaction_latency(w3, tx_hash)
            receipt = w3.eth.wait_for_transaction_receipt(tx_hash)
            transaction_hashes[node] = receipt.transactionHash.hex()
            block_number = receipt.blockNumber


# Display transaction details
print(f"Anomaly reported from Node {node}: Data {data} ({node_type})")
print(f"Transaction: {receipt.transactionHash.hex()}")
print(f"Block Number: {block_number}")
print(f"Gas used: {receipt.gasUsed}")
print(f"Transaction Latency: {latency:.4f} seconds")


# Notify other nodes about the anomaly
notify_other_nodes(node, data, "Anomaly", contract, receipt.transactionHash.hex())


# Isolate the node and mark it for removal
isolate_node(contract, node, w3)

```

```

        nodes_to_remove.add(node)

        print(f"Anomaly isolated and notified for Node {node}\n")
    except Exception as e:
        print(f"Error handling anomaly for Node {node}: {e}")

# Handle potential DDoS attacks
for node in list(ddos_attacks.keys()):
    data = ddos_attacks[node]
    node_type = check_node_type(node, nodes) # Check if the node is an insider or external
    print(f"Potential DDoS detected from Node {node}: Data {data}\n")
    try:
        # Report the DDoS to the smart contract and measure latency
        tx_hash = contract.functions.reportDDoS(data).transact({'from': w3.eth.default_account})
        latency = measure_transaction_latency(w3, tx_hash)
        receipt = w3.eth.wait_for_transaction_receipt(tx_hash)
        transaction_hashes[node] = receipt.transactionHash.hex()
        block_number = receipt.blockNumber

        # Display transaction details
        print(f"Potential DDoS reported from Node {node}: Data {data} ({node_type})")
        print(f"Transaction: {receipt.transactionHash.hex()}")
        print(f"Block Number: {block_number}")
        print(f"Gas used: {receipt.gasUsed}")
        print(f"Transaction Latency: {latency:.4f} seconds")

        # Notify other nodes about the DDoS attack
        notify_other_nodes(node, data, "DDoS", contract, receipt.transactionHash.hex())

        # Isolate the node and mark it for removal
        isolate_node(contract, node, w3)
        nodes_to_remove.add(node)

```

```

        print(f"DDoS isolated and notified for Node {node}\n")
    except Exception as e:
        print(f"Error handling DDoS for Node {node}: {e}")

# Remove compromised nodes from the network
nodes_removed = {}
for node in nodes_to_remove:
    try:
        channel = node_channels[node] # Get the channel associated with the node
        remove_node_from_network(node, channel) # Remove the node from the network
        nodes_removed[node] = channel # Record the removed node
        print(f"Removed Node {node} with channel {channel} from the network.\n")
    except Exception as e:
        print(f"Error removing Node {node} from network: {e}")

# Reinitialize the contract with non-compromised nodes
non_compromised_nodes = [node for node in normal_nodes if node not in nodes_to_remove]
if non_compromised_nodes:
    try:
        reinitialize_contract(w3, non_compromised_nodes) # Reinitialize the contract
        print(f"Contract is reinitialized with non-compromised nodes.\n")
    except Exception as e:
        print(f"Error reinitializing contract: {e}")

# Display the final list of non-compromised nodes
print("Non-compromised nodes after detection:\n")
for node in non_compromised_nodes:
    print(f"Node {node} with channel {node_channels[node]}")

# Stop memory tracking and measure resource usage
end_time = time.time() - start_time

```

```

current, peak = tracemalloc.get_traced_memory()
tracemalloc.stop()
cpu_usage = measure_resource_usage()

# Display resource usage and timing information
print(f"Total turn around time {end_time:.6f}s")
print(f"Memory Usage {peak/1024:.2f}KB")

# Main execution block
if __name__ == "__main__":
    detect_anomalies_and_ddos() # Run the detection process

```

2. AnomalyDetection.sol

```

// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract AnomalyDetection {
    address public owner;
    uint public anomalyThreshold;

    mapping(address => bool) public isolatedNodes;
    mapping(address => bool) public compromisedNodes;
    address[] public normalNodes;

    constructor(address[] memory _normalNodes, uint _threshold) {
        owner = msg.sender;
        normalNodes = _normalNodes;
        anomalyThreshold = _threshold;
    }
}

```

```

function setAnomalyThreshold(uint _threshold) public {
    require(msg.sender == owner, "Only owner can set the threshold");
    anomalyThreshold = _threshold;
}

function detectAnomaly(uint data) public {
    require(data > anomalyThreshold, "Data does not exceed threshold");
    compromisedNodes[msg.sender] = true;
}

function isolateNode(address node) public {
    require(msg.sender == owner, "Only owner can isolate nodes");
    isolatedNodes[node] = true;
}

function reportDDoS(uint data) public {
    require(data > anomalyThreshold, "Data does not exceed threshold for DDoS");
    compromisedNodes[msg.sender] = true;
}

function getNormalNodes() public view returns (address[] memory) {
    return normalNodes;
}

function getCompromisedNodes() public view returns (address[] memory) {
    uint length = 0;
    for (uint i = 0; i < normalNodes.length; i++) {
        if (compromisedNodes[normalNodes[i]]) {
            length++;
        }
    }
}

```

```

address[] memory compromised = new address[](length);
uint index = 0;
for (uint i = 0; i < normalNodes.length; i++) {
    if (compromisedNodes[normalNodes[i]]) {
        compromised[index] = normalNodes[i];
        index++;
    }
}
return compromised;
}
}

```

3. simulate_can.py

```

import can
import time
import threading

def simulate_can_network(channel, node_id, anomaly_delay=None, ddos_interval=None):
    bus = can.interface.Bus(channel=channel, interface='socketcan')
    start_time = time.time()
    ddos_start_time = time.time() if ddos_interval else None

    print(f"Node {node_id} started sending messages on {channel}")

    while True:
        current_time = time.time()
        node_id_int = ord(node_id[0]) if isinstance(node_id, str) else node_id

        if node_id == 'Node1':
            # Node 1 can send both normal and anomalous data and DDoS traffic
            if anomaly_delay and (current_time - start_time) > anomaly_delay:

```

```

    data = [node_id_int, 0xFF, 0xFF, 0xFF] # Anomalous data
    print(f"Node {node_id} switched to anomalous data")
else:
    data = [node_id_int, 0x00, 0x00, 0x00] # Normal data

if ddos_interval and (current_time - start_time) > anomaly_delay:
    for _ in range(10): # Send a DDoS attack with 10 messages
        try:
            msg = can.Message(arbitration_id=0x123, data=data, is_extended_id=False)
            bus.send(msg)
            print(f"Node {node_id} sent DDoS message: {msg}")
        except can.CanError as e:
            print(f"Node {node_id} failed to send DDoS message: {e}")
        time.sleep(0.1) # Short interval between DDoS messages
    else:
        try:
            msg = can.Message(arbitration_id=0x123, data=data, is_extended_id=False)
            bus.send(msg)
            print(f"Node {node_id} successfully sent message: {msg}")
        except can.CanError as e:
            print(f"Node {node_id} failed to send message: {e}")

    time.sleep(1) # Normal interval between messages
else:
    # Node 2, Node 3, and Master Node only send normal traffic
    data = [node_id_int, 0x00, 0x00, 0x00] # Normal data
    try:
        msg = can.Message(arbitration_id=0x123, data=data, is_extended_id=False)
        bus.send(msg)
        print(f"Node {node_id} successfully sent message: {msg}")
    except can.CanError as e:

```

```

        print(f"Node {node_id} failed to send message: {e}")
    time.sleep(1) # Normal interval between messages

if __name__ == "__main__":
    nodes = [
        ('MasterNode', 'vcan0', None, None), # Master Node
        ('Node1', 'vcan1', 20, 10), # Anomaly after 20 seconds, DDoS interval of 10 seconds
        ('Node2', 'vcan2', None, None),
        ('Node3', 'vcan3', None, None)
    ]

    for node_id, channel, anomaly_delay, ddos_interval in nodes:
        threading.Thread(target=simulate_can_network, args=(channel, node_id, anomaly_delay,
ddos_interval)).start()

```

Appendix B – Ethical Approval mail

SRIDHAR, VISHNU PRASHANTH (PGT)

From: WMG-FTMasters, Resource
Sent: 16 May 2024 16:51
To: SRIDHAR, VISHNU PRASHANTH (PGT)
Subject: Ethical Approval waived for this research

Date of Approval: **16th May 2024**

Dear **Vishnu Prashanth Sridhar**

Warwick ID number: **5569451**

Project: **A blockchain-Enabled Evaluator of Indicators of Compromise in Connected Vehicles**

Your ethics application number is **WMG-2024-FTMSc--R_7KpctVaDucxNQ6S**

Your supervisor **Dr. Gregory Epiphaniou** has recommended to the Cyber Ethics Panel the following outcome: **Ethical approval to be waived for this research.**

The Cyber Ethics Panel has confirmed this outcome.

You are reminded that you must now adhere to the answers and detail given in the completed ethics form. If anything changes in your research such that any of your answers change to the form for which you received an **ethical waiver** for, then you must contact your supervisor to check if you need to reapply for or update your **ethical waiver** before you proceed with data collection.

When you submit your dissertation, please write **N/A against the ethical approval field** on the cover page of the submission and include a copy of this email into the Appendices of your dissertation.

Kind regards,

WMG Projects Team

Appendix C – Ethical Approval Courses Completion

From: Do not reply to this email (via Moodle@Warwick) <no-reply-moodle@warwick.ac.uk>

Sent: Wednesday, April 10, 2024 11:29 PM

To: SRIDHAR, VISHNU PRASHANTH (PGT) <Vishnu-Prashanth.Sridhar@warwick.ac.uk>

Subject: Course completed

Congratulations!

You have completed the course [Information Security Smart](#).



Certificate

Number: 2213170291

This is to certify that

Vishnu Prashanth Sridhar
of University of Warwick

Successfully completed the course
Research Integrity: Concise (core course)

as part of the Epigeum Online Course System with a score of 96%.

Dated: 26 March 2024

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